Education Subsidies as Insurance against Income Risk in Fertility Decisions*

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Abstract

Empirical studies suggest that income uncertainty depresses fertility, yet its policy implications remain underexplored. This paper shows that income-tested education subsidies partially insure households against income risk in fertility decisions. I build and calibrate an incomplete-market general-equilibrium lifecycle model with fertility and college enrollment choices, in which altruistic parents make asset transfers to support their children's college education. Having more children increases the risk that parents may be unable to finance their children's education after bad income shocks or, if they do finance it, must endure substantially lower lifetime consumption. This mechanism discourages fertility under income uncertainty and generates the observed negative relationship between income risk and fertility. Counterfactual experiments show that income-tested college subsidies insure households against these risks by reducing education costs contingent on bad income shocks. The resulting fertility responses amplify the aggregate effects of the subsidy, including increases in educational attainment and intergenerational mobility.

Keywords: Fertility, incomplete market, college subsidy, income risk
JEL codes: C68, I28, J13, J24

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1 Introduction

Total fertility rates have reached historic lows, contributing to population decline and a shrinking share of the working-age population. Persistently low fertility therefore raises important macroeconomic concerns, including labor force contraction, pressures on the sustainability of social security systems, and lower living standards. Among the determinants of fertility, labor market factors—such as job flexibility and income uncertainty—have become increasingly important in explaining cross-country variation in fertility (Doepke et al., 2023). In particular, empirical studies consistently document a negative relationship between measures of perceived income uncertainty and fertility (e.g., Chabé-Ferret and Gobbi, 2018; Alderotti et al., 2021). Despite the broad consensus that income uncertainty is an important determinant of fertility choices, little is known about its policy implications and the broader economic consequences of such policies.

This paper addresses this gap and makes three main contributions. First, I show analytically and quantitatively that income-tested college subsidies, prevalent across countries, act as partial insurance against income risk in fertility decisions. Having more children increases the risk that parents may be unable to finance their children's education after bad income shocks or, if they do finance it, must endure substantially lower lifetime consumption. Income-tested college subsidies insure households against these risks by reducing education costs contingent on bad income shocks. Second, I show that the resulting fertility responses amplify the aggregate effects of college subsidies, including increases in average educational attainment and intergenerational mobility. Third, this study is the first to develop a general equilibrium model that jointly incorporates fertility and college enrollment choices, thereby allowing us to quantify the insurance role of college subsidies in fertility decisions and their aggregate implications.

I begin with a simple analytical model of fertility choice under uninsurable income risk. In this model, ex-ante identical households make fertility choices under income uncertainty, with spending on children assumed to be fixed and exogenously given, and income levels are realized after fertility choices are made. I first show that a mean-preserving spread of the income distribution (i.e., a rise in income uncertainty) in the model leads to lower fertility. The intuition is that having children can become very costly in utility terms following a bad income realization. A given negative income shock results in substantially lower consumption when households have more children than when they have fewer children (or lower spending on children if such spending were endogenous), thereby raising marginal utility. However, parents cannot "reduce" the number of children after facing a bad income shock because fertility choices are irreversible. Therefore, income uncertainty may deter parents from having another child, lowering fertility rates. I then consider an income-tested subsidy that redistributes income from households with positive (good) income shocks to those with negative (bad) shocks, ensuring that the expected value of the subsidy received is zero. This design rules out fertility effects arising from lower expected monetary costs of children (or the lower "price" of children), which I refer to as the *price effect*. I show that despite the absence of the price effect, income-tested subsidies still increase fertility: they reduce the dispersion in marginal utilities across income states conditional on having children, thereby lowering the expected marginal costs of children, even though the expected monetary costs remain unchanged. I refer to this fertility increase as the *insurance effect* of the subsidy.

Building on these theoretical predictions, I next evaluate the quantitative importance of this insurance effect relative to other potential channels, as well as the macroeconomic implications of the resulting fertility responses. To do so, I develop an incomplete-market general-equilibrium lifecycle model with several key elements: (1) fertility choices, (2) college enrollment choices, (3) intergenerational linkages between parents and children through asset transfers and exogenous transmission of innate abilities and tastes for education, and (4) imperfect substitutability between college and non-college graduate workers in the aggregate production function. The model is calibrated using data from the Japanese Panel Survey of Consumers (JPSC), focusing on households in which women were born between 1959 and 1969. The calibrated model replicates key cross-sectional moments across the income distribution, including fertility, expenditures on children prior to college enrollment, parental transfers to college students, college enrollment rates, and intergenerational persistence in income and education. I also estimate the stochastic component of the income process, which follows a first-order autoregressive process.

Japan provides a useful setting for this analysis. College enrollment rates in Japan have increased steadily over time, with current enrollment rates approaching 60%, and more than 80% of parents report wanting their children to attend college (Figure D.4). In practice, parents can typically send their children to college if they can afford the associated costs, since college capacity is generally not a binding constraint.[1] However, higher education in Japan is costly and receives relatively limited public financial support compared with many other developed countries (Figure D.1). Annual college expenditures—mainly tuition and related expenses—amount to nearly half of median annual household income,[2] and parents finance more than 60% of these costs on average. In response, the Japanese government introduced an income-tested college subsidy in 2020 as part of a pronatalist policy package and has expanded its coverage in subsequent years.[3] The model is used to quantify the effects of this policy, including the insurance channel highlighted above.

Before turning to the policy analysis, I first quantify the role of income uncertainty in shaping fertility. To do so, I eliminate income risk by removing the stochastic component from the wage process. This increases completed fertility by 15.8% on average. The effect is sizable; a 15.8% increase would bring Japan's 2024 fertility rate (1.14), the sixth lowest among OECD countries, much closer to the OECD average (1.40). The positive effect is particularly pronounced among lower-income households, whose ability to self-insure against bad income shocks is relatively limited.

I then examine the effects of the income-tested college subsidies currently in place in Japan, which are absent from the baseline economy. The current grant scheme provides cash transfers to college students from households in approximately the bottom 40% of the income distribution, and for eligible students, the full grant covers roughly two-thirds of average college expenditures.[4] I incorporate into the students' budget constraint a

[1] For example, a substantial share of private universities fail to meet their enrollment targets; see reports by the Japan Private School Promotion and Mutual Aid Corporation (https://www.shigaku.go.jp/files/shigandoukouR5.pdf, in Japanese).

[2] The median household income in 2021 was 4.4 million yen, according to the Comprehensive Survey of Living Conditions conducted by the Ministry of Health, Labour and Welfare (MHLW).

[3] See Online Appendix D for details.

[4] While the policy initially targeted students from households at the lower end of the income distribution when it was introduced in 2020, its coverage has expanded gradually in recent years. This model does not allow students to access financial aid (grants or loans) provided by institutions besides the govern-

grant function that approximates the current subsidy schedule. I find that introducing the income-tested subsidy—holding prices, tax rates, and the household distribution constant—leads to a 5.7% increase in aggregate fertility. However, the responses differ markedly across households, even qualitatively: fertility rises by 17.2% among college graduates but falls by 10.2% among high-school graduates.

To understand these responses, I decompose the direct effect into a price effect and an insurance effect. The price effect is obtained by solving households' decision problems in an environment in which households anticipate with certainty, at the time fertility decisions are made, that their children will receive the expected value of college grants. It therefore captures the fertility effects arising from lower expected education costs, abstracting from the insurance value of the subsidy that becomes available when households experience bad income realizations. The insurance effect, in turn, is measured as the difference between the fertility response under the income-tested subsidy schedule and that under a uniform subsidy that provides the same expected subsidy amount with certainty. It therefore captures the fertility effects arising from the insurance value of state-contingent educational support, holding expected education costs constant.

The decomposition reveals that the insurance effect is the primary driver of the positive aggregate fertility response, whereas the price effect is the primary source of the heterogeneous fertility responses across education groups. Less-educated households, whose children are less likely to attend college in the benchmark economy, are more likely to send their children to college as a result of the lower education costs. Nevertheless, college attendance remains more costly than non-attendance because the grant does not fully cover educational expenses. Consequently, the subsidy raises the expected educational investment per child and makes childrearing effectively more expensive, reducing fertility. By contrast, children of college-educated parents already have a high probability of attending college in the benchmark economy. For these households, the subsidy primarily lowers the expected cost of children and generates a positive income effect, thereby increasing fertility. Most importantly, the insurance effect is positive and sizable for both education groups. While the price effect alone reduces aggregate fertility because of the quantity–quality trade-off faced by less educated households, the insurance effect is sufficiently strong to more than offset this force, resulting in a positive overall impact on fertility.

I then solve the model in general equilibrium, allowing factor prices, tax rates, and the household distribution to adjust. I find that the fertility responses amplify a number of the policy's key outcomes, such as college enrollment. As described above, the increase in fertility is particularly pronounced among college-graduate parents, whose children are more likely to attend college. Given the intergenerational persistence of abilities, preferences, and assets, the additional children born as a result of the policy are themselves more likely to enroll in college. Consequently, the policy generates an endogenous compositional shift toward households with a higher propensity to invest in higher education, further increasing college enrollment in the long run. I demonstrate the importance of this fertility channel by tracing the policy's effects along the transition path and by comparing the long-run results with those obtained under an exogenous fertility setup.

ment (e.g., colleges or other private entities). In 2018, among students in four-year universities who used any financial aid, 83.8% used only government-provided aid, 8.6% used only non-government-provided aid, and 7.6% used both, according to the Student Life Survey (SLS, 2018).

2 Related Literature

This paper contributes to several strands of literature. The first is the macroeconomic analysis of college education subsidies.[5] The literature highlights important implications of the college subsidies, including the GE feedback through an increased supply of college-educated workers (e.g., Krueger and Ludwig, 2016), the crowding-out of parental transfers and student labor supply that may dampen policy effects (Abbott et al., 2019), and the complementarity between college subsidies and policies promoting pre-college human capital accumulation (Krueger, Ludwig, and Popova, 2025). I contribute to this literature by incorporating fertility choices into an otherwise standard framework—an incomplete-markets, GE lifecycle model with college enrollment and intergenerational linkages. The extended framework highlights a new channel through which college subsidies operate: their insurance role in fertility decisions and the resulting amplification of policy effects.

Second, this paper builds on macroeconomic studies based on the quantity-quality trade-off framework.[6] In particular, this study is closely related to De la Croix and Doepke (2004), Kim, Tertilt, and Yum (2024), and Zhou (2025), which examine the macroeconomic implications of education policies in models with endogenous fertility and parental investments. The key contribution of this study is to demonstrate the insurance role of income-tested education subsidies in fertility choices. From a modeling perspective, the closest paper is Daruich and Kozlowski (2020). They develop a partial equilibrium lifecycle model with college enrollment, fertility, and inter-vivos transfers to investigate the role of fertility choices and family transfers in explaining intergenerational mobility in the U.S. This study extends their framework by adopting a GE approach to examine the macroeconomic effects of education policies.

Third, this study is also related to the literature on fertility choices in incomplete-market models and, more broadly, to the literature on the implications of heterogeneous family structures under incomplete markets.[7] Previous studies demonstrate that having a child can be considered making "consumption commitments," and that income volatility thus makes households hesitate to have children, leading to a lower fertility (Santos and Weiss, 2016; Sommer, 2016).[8] This study contributes to the literature by examining policies that can potentially insure against income risks in fertility decisions and highlights the insurance effects of income-tested college subsidies.

Finally, this study contributes to the literature on pro-natal policies based on structural models.[9] My contribution is to study the effects of college subsidies, a novel pro-

[5] For example, Krueger and Ludwig (2016); Abbott, Gallipoli, Meghir, and Violante (2019); Matsuda and Mazur (2022); Krueger, Ludwig, and Popova (2025).

[6] For example, De La Croix and Doepke (2003); De la Croix and Doepke (2004); Manuelli and Seshadri (2009); Greenwood, Guner, Kocharkov, and Santos (2016); Cordoba, Liu, and Ripoll (2019); Daruich and Kozlowski (2020); Zhou (2025); Kim, Tertilt, and Yum (2024); Kitao and Nakakuni (2024).

[7] For example, Cubeddu and Ríos-Rull (2003); Schoonbroodt and Tertilt (2014); Vandenbroucke (2014); Santos and Weiss (2016); Sommer (2016); Bacher, Grübener, and Nord (2024); Coskun and Dalgic (2024); Guner, Kaya, and Sánchez-Marcos (2024); Adamopoulou, Hannusch, Kopecky, and Obermeier (2025).

[8] This effect is particularly pronounced in the early stages of life, leading to delayed fertility. Since the ability to reproduce declines with age, delayed fertility ultimately results in lower fertility rates (Sommer, 2016). Similarly, Santos and Weiss (2016) show that an increase in income volatility leads to delays in marriage, as marriage often entails a consumption commitment in the form of raising children.

[9] Previous studies examine the effects of childcare subsidization (e.g., Bick, 2016), cash transfers (e.g., Kim, Tertilt, and Yum, 2024; Nakakuni, 2024), both of them (e.g., Zhou, 2025; Hagiwara, 2025), parental

natal policy implemented in Japan, and to demonstrate that the insurance channel is an important mechanism through which such policies can affect fertility choices.

3 Analytical Model

This section develops a simple analytical model of fertility choice under uninsurable income risk. I derive two main results: (1) greater income uncertainty (a mean-preserving spread of the income distribution) reduces fertility; and (2) an income-tested subsidy increases fertility in the presence of income risk.

Setup: Consider a decision problem in which ex-ante identical households choose the number of children n to maximize expected utility, facing uninsurable income risk. For simplicity, n is assumed to be continuous and strictly positive. After the fertility choice is made, income I is realized from a distribution with cumulative distribution function $F(I)$ and support $[I_{min}, I_{max}]$, where $0 < I_{min} < I_{max} < \infty$. Each child entails a fixed monetary cost $\underline{a} > 0$, net of a government subsidy $\bar{g}(I)$, which I allow to depend on income (i.e., the subsidy is income-dependent or income-tested). The resulting net cost per child, denoted by $a(I)$, is given by $a(I) = \underline{a} - \bar{g}(I)$, where $a(I) \in (0, \infty)$ for all I. Households derive utility from both the number of children and consumption c, where consumption equals income net of total child-rearing costs, that is, $c = I - na(I)$. Let $\mathcal{N}$ denote the feasible set of fertility choices ensuring positive consumption for all income realizations, that is, $\mathcal{N} = \{n \mid n > 0,\ I - na(I) > 0\ \forall I \in [I_{\min}, I_{\max}]\}$. Also, let $b(\cdot)$ denote the utility function for the number of children and $V(\cdot)$ the utility function for consumption. The fertility choice problem is given by:

$$\max_{n \in \mathcal{N}} \left\{ b(n) + \int_{I_{\min}}^{I_{\max}} V\left(I - na(I)\right) dF(I) \right\}.$$

The utility functions $b(\cdot)$ and $V(\cdot)$ satisfy the standard properties: $b' > 0$, $b'' < 0$, $\lim_{n \to 0^+} b'(n) = \infty$, $V' > 0$, $V'' < 0$, $V''' > 0$, and $\lim_{c \to 0^+} V'(c) = \infty$.

Effects of Income Risk on Fertility: I first show that an increase in income risk—in the form of a mean-preserving spread of the income distribution $F(I)$—reduces fertility n. To this end, I consider a symmetric two-point distribution for income. Specifically, income can take either a high or a low value, each occurring with probability 0.5:

$$I = \begin{cases} \bar{I} + \nu & \text{with probability } 0.5, \\ \bar{I} - \nu & \text{with probability } 0.5. \end{cases}$$

Here, $\bar{I} > 0$ is the mean of the income distribution, and $\nu < \bar{I}$ is a parameter that governs the income dispersion. Also, because the focus is now on the pure effect of income uncertainty on fertility, I do not consider the income-tested subsidy, that is, $\bar{g}(I) = 0$ for each I. Given this setup, the following proposition holds.

leave policies (e.g., Erosa, Fuster, and Restuccia, 2010; Yamaguchi, 2019; Wang, 2022; Kim and Yum, 2023), and tax reform (e.g., Jakobsen, Jørgensen, and Low, 2022).

Proposition 1. *Let $n(\nu)$ be the optimal number of children, represented as a function of the income dispersion parameter ν. It holds that*

$$\frac{dn}{d\nu} < 0. \tag{1}$$

Namely, a mean-preserving spread of the income distribution (i.e., an increase in income risk) reduces fertility.

Proof. The optimization problem is specified as

$$\max_{n \in \mathcal{N}} \left\{ b(n) + \frac{1}{2} V(\bar{I} - \nu - n\underline{a}) + \frac{1}{2} V(\bar{I} + \nu - n\underline{a}) \right\}.$$

The first-order condition implies that the optimal number of children, denoted by $n(\nu)$, satisfies

$$\underline{a}^{-1} b'(n(\nu)) = \frac{1}{2} \Big[V'(\bar{I} + \nu - n(\nu)\underline{a}) + V'(\bar{I} - \nu - n(\nu)\underline{a}) \Big]. \tag{2}$$

Next, I take derivatives of both sides of equation (2) with respect to ν:

$$\underline{a}^{-1} b''(n(\nu)) \frac{dn}{d\nu} = \frac{1}{2} \left[V''(c_h) - V''(c_l) - (V''(c_h) + V''(c_l)) \cdot \frac{dn}{d\nu} \cdot \underline{a} \right],$$

where c_h and c_l represent the consumption levels when the realized income is high and low, that is, $c_h = \bar{I} + \nu - \underline{a}n(\nu)$ and $c_l = \bar{I} - \nu - \underline{a}n(\nu)$. After rearranging the terms and solving for $dn/d\nu$, I obtain the stated result (1):

$$\frac{dn}{d\nu} = \frac{\overbrace{V''(c_h) - V''(c_l)}^{>0 \text{ (since } V'''>0 \text{ and } c_h > c_l)}}{\underbrace{2\underline{a}^{-1} b''(n) + \underline{a}[V''(c_h) + V''(c_l)])}_{<0 \text{ (since } b''<0 \text{ and } V''<0)}} < 0$$

□

The intuition and mechanism are analogous to the reduction in consumption caused by precautionary saving. The key assumption underlying this proposition is that the third derivative of utility with respect to consumption is positive, which is also the central condition that generates precautionary saving (Leland, 1968; Kimball, 1990). When households choose the number of children, they do not yet know which income level will be realized once the children are born. If they increase fertility marginally, the marginal utility of consumption becomes particularly high in states where income turns out to be low. This is because fertility is an irreversible choice—they cannot reduce the number of children after observing a bad income realization—and they must allocate resources to their children even when income is low. Consequently, households are cautious about having "too many" children when making fertility decisions under income uncertainty. This precaution leads them to limit fertility, thereby reducing the number of children they choose to have.[10]

[10] While this theoretical prediction is broadly consistent with the empirical literature (e.g., Chabé-Ferret

Effects of Income-tested Subsidies on Fertility under Income Uncertainty: Now I study the effects of income-tested subsidies on fertility under income uncertainty. Here, I consider a general income distribution $F(I)$ with support $[I_{\min}, I_{\max}]$, where $\bar{I} = \mathbb{E}[I]$ denotes mean income. I introduce an income-tested subsidy of the form

$$\bar{g}(I;\varepsilon) = (\bar{I} - I) \cdot \varepsilon,$$

where $\varepsilon \geq 0$ is a policy parameter that governs the degree of the redistribution. Note that this subsidy function $\bar{g}(I;\varepsilon)$ implies that (1) the subsidy is income-contingent (or income-tested), where households with below-average income realizations receive a positive subsidy, while those with above-average income realizations receive a negative subsidy or incur a "tax," and (2) the expected cost of children remain constant at $\underline{a}$ because the expected value of the subsidy is zero for any ε (i.e., $\mathbb{E}\bar{g}(I;\varepsilon) = \mathbb{E}(\bar{I} - I) \cdot \varepsilon = 0$). With this new notation, the unit cost of children is now denoted by $a(I;\varepsilon) = \underline{a} - \bar{g}(I;\varepsilon)$. For convenience, I express the optimal fertility choice as a function of the redistribution parameter, $n(\varepsilon)$. Similarly, I express the realized consumption as a function of the realized income I and the redistribution parameter ε, $c(I;\varepsilon) = I - n(\varepsilon)a(I;\varepsilon)$.

I define the parameter space for ε as

$$\mathcal{E} \equiv \{\varepsilon \mid 1 - \varepsilon \cdot n(\varepsilon) > 0,\ \varepsilon \geq 0\},$$

which ensures that higher (lower) income realizations correspond to higher (lower) consumption (i.e., $\partial c(I;\varepsilon)/\partial I > 0$). This rules out excessively aggressive redistribution schemes under which a household with higher pre-redistribution income would end up with lower post-redistribution income than a poorer household.

The first-order condition implies that the optimal fertility choice is characterized as:

$$b'(n(\varepsilon)) = \mathbb{E}[a(I;\varepsilon) \cdot V'(c(I;\varepsilon))] \tag{3}$$

This equates the marginal utility from the number of children to the expected marginal costs of having children. Since the marginal utility from children decreases (since $b'' < 0$), this equation implies that the fertility decreases with the expected marginal costs of having children, which comprise both the monetary costs of children and the utility costs associated with foregone consumption. When they make fertility choices, the marginal costs of having children are uncertain because the amount of subsidy received and consumption level are uncertain; either source of uncertainty is income risk.

Note that the first-order derivative of the optimal fertility choice n with respect to ε measures how fertility responds to an increase in the extent of redistribution embedded in income-tested subsidies. In particular, evaluating this derivative at $\varepsilon = 0$ captures the fertility response to the introduction of an income-tested subsidy, since $\varepsilon = 0$ represents

and Gobbi, 2018; De Paola et al., 2021), it is worth noting that there is also a theoretical possibility of a positive relationship between income uncertainty and fertility. For example, in environments where children serve as a form of old-age insurance and parents rely on them for future support (Boldrin and Jones, 2002), higher income uncertainty may instead increase fertility, as households can partially insure against income risk by having more children and relying on intergenerational transfers in old age. This mechanism is likely to be less important in high-income countries such as Japan, where public pension systems and well-developed financial markets substantially reduce households' reliance on children as a source of old-age support.

a scenario with no subsidy. I obtain the following result.

Proposition 2. *Let $n(\varepsilon)$ be the optimal number of children given the redistribution policy parameter $\varepsilon \in \mathcal{E}$. It holds that:*

$$n'(0) \equiv \left.\frac{dn}{d\varepsilon}\right|_{\varepsilon=0} > 0,$$

which means that the introduction of an income-tested subsidy increases fertility.

Proof. First, differentiate the optimality condition for fertility (3),

$$b'(n(\varepsilon)) = \mathbb{E}[a(I;\varepsilon) \cdot V'(c(I;\varepsilon))],$$

with respect to ε. Evaluating the differentiated equation at $\varepsilon = 0$ yields an equation containing $n'(0)$. Rearranging terms and solving for $n'(0)$ establishes the proposition. The detailed derivation is provided in Appendix A.1. □

The intuition for why the introduction of an income-tested subsidy increases fertility is as follows. The previous proposition shows that income uncertainty reduces fertility because having more children makes low-income states more costly in utility terms, discouraging households from having more children under income uncertainty. An income-tested subsidy mitigates this risk by providing transfers in low-income states. This insurance effect reduces the expected *utility cost* of children ($\mathbb{E}[a(I;\varepsilon)V'(c(I;\varepsilon))]$)—even though the expected *monetary cost* of children remains constant ($\mathbb{E}[a(I;\varepsilon)] = \underline{a}$)—by providing insurance against states in which consumption would otherwise be particularly low. While the subsidy imposes a cost in high-income states—raising the marginal utility of consumption relative to the no-subsidy case—the benefit from insurance in low-income states dominates. In particular, when marginal utility is convex (i.e., $V''' > 0$), the reduction in marginal utility in bad states outweighs the increase in marginal utility in good states. As a result, the expected utility cost of having children declines, leading to higher fertility.

Toward the quantitative model: This section presents a simple analytical model showing that income uncertainty reduces fertility, while income-tested subsidies can increase fertility by providing insurance against income risk. These findings raise several further questions: (1) to what extent does income risk depress fertility in practice? (2) how effectively do real-world income-tested education subsidies provide such insurance? and (3) what are the implications of the resulting fertility responses for broader macroeconomic outcomes, such as aggregate output and income distribution? The current model is not designed to answer these questions because it abstracts from several important features. These include (i) a more realistic lifecycle income process, (ii) savings and labor supply decisions that provide self-insurance against income risk, (iii) endogenous expenditures on children, and (iv) market production and general equilibrium effects, which are necessary for studying the policy's macroeconomic implications. The next section therefore develops a quantitative macroeconomic model that incorporates these features.

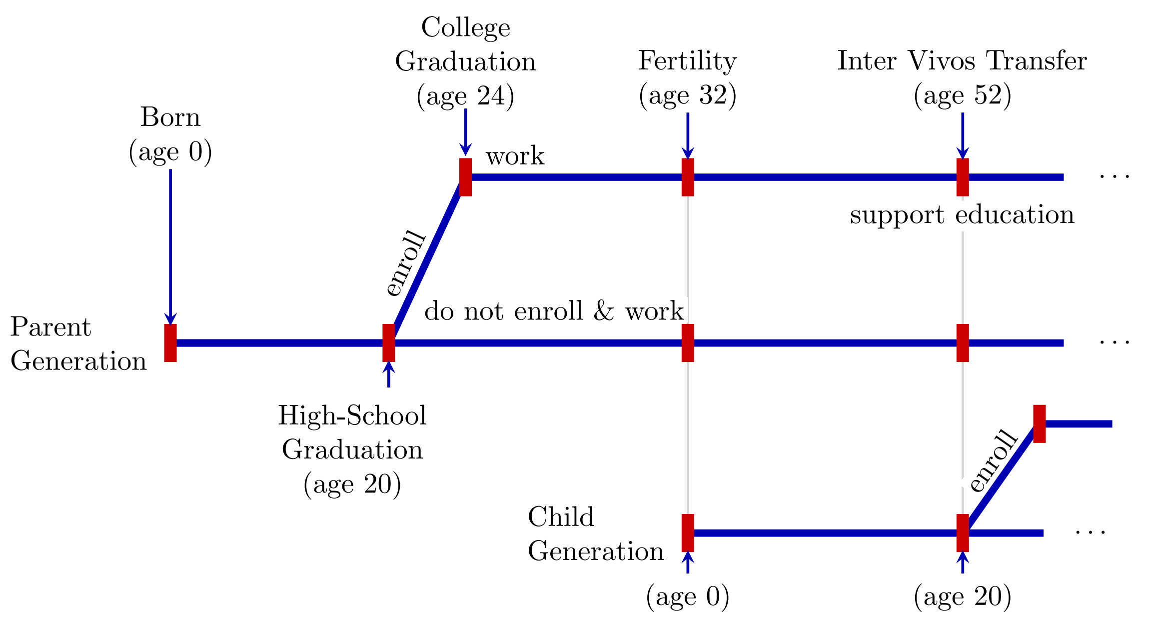
4 Quantitative Model

I incorporate fertility choices into an otherwise standard framework in the macroeconomic literature of college subsidies: an incomplete-market general-equilibrium (GE) lifecycle model with college enrollment choices and intergenerational linkages. Section 4.1 provides an overview of the lifecycle of this economy. Section 4.2 elaborates on the primitives of the model, and then Section 4.3 formulates the household's problems. Section 4.4 discusses the equilibrium of the economy.

4.1 Overview of the lifecycle

Figure 1 illustrates the key stages—events and choices—in the household lifecycle in the model. Time is discrete, and one period corresponds to four years. Let j denote the agent's age. They live with their parents until they graduate from high school at the beginning of age $j = J_E$ (biological age 20); until then, they make no decisions. After graduating high school, they choose whether to attend college. This decision is represented by the node labeled "High-School Graduation" in the figure. If they do not enroll in college, they enter the labor market as high-school graduates. If they choose to attend college, it takes four years (one model period) to complete. Upon graduation, they enter the labor market as college graduates at age $J_E + 1$ (biological age 24). This model does not consider the possibility of college dropout, as the dropout rate is negligible in Japan.[11]

Figure 1: Key stages in the model's lifecycle



At the beginning of age J_F (biological age 32), they make fertility choices by choosing how many children they have, $n \in \{0, 1, 2, 3, 4\}$.[12] The timing of this decision is the

[11] For example, the dropout rate was 2.5% in 2021, according to the Ministry of Education, Culture, Sports, Science and Technology.

[12] I discuss this fixed fertility timing assumption in Online Appendix E.3.

same for both high school and college graduates.[13] If they have children, their children's lifecycle starts at this point, represented by the node labeled "Child Generation" in the figure. After their children graduate from high school, corresponding to the beginning of the parent's age J_{IVT} (biological age 52), they decide how much money to transfer to their children. This inter vivos transfer (IVT) decision affects the children's college enrollment choices. Households retire from the labor market at the beginning of age J_R (biological age 64). After that period, they face mortality risks. Every period, a certain fraction of them dies because of exogenous mortality shocks and exits the economy. They can live for at most J years (until biological age 103), at which point they die with certainty. To provide an overview of the household's problem, Table 1 summarizes the choice and state variables, together with the age or period at which each variable is relevant. Each variable is explained in detail in the following subsections.

Table 1: Summary of the choice and state variables over the lifecycle

Description	**Notation**	**Age/Period**
Choice variables		
Consumption	$c \in \mathbb{R}_{++}$	20–103
Leisure	$l \in [0, 1]$	20–103
Asset position	$a' \in [\underline{A}, \infty)$	20–103
Education: high-school (=HS) or college (=CL)	$e \in \{HS, CL\}$	20
Number of children (fertility)	$n \in \{0, 1, 2, 3, 4\}$	32
Expenditures on children	$q \in \mathbb{R}_{++}$	32–51
Inter vivos transfer	$a_{IVT} \in \mathbb{R}_{+}$	52
State variables		
Age	$j \in \{J_E, ..., J\}$	20–103
Asset position	$a \in [\underline{A}, \infty)$	20–103
Innate ability	$h \in \mathbb{R}_{++}$	20–63
Psychic costs of college education	$\phi \in \mathbb{R}_{+}$	20
Household income	$I \in \mathbb{R}_{++}$	20
Education: high-school (=HS) or college (=CL)	$e \in \{HS, CL\}$	20–63
Stochastic labor productivity	$z \in \mathbb{R}$	20–63
Number of children	$n \in \{0, 1, 2, 3, 4\}$	32–51
Children's innate ability	$h_k \in \mathbb{R}_{++}$	52
Children's psychic costs of college education	$\phi_k \in \mathbb{R}_{+}$	52

Note: The "Age/Period" column refers to biological age at which each variable is relevant. The borrowing limit, $\underline{A}$, varies over the lifecycle, as discussed in the main text.

[13] In the data, differences in the average timing of first birth are not substantial. In my JPSC sample, the average age at first birth is 29.75 for high school graduates and 31.34 for college graduates.

4.2 Economic Environment

Demographics: Every period, a mass of new cohorts enters the economy. The size of the new cohort is endogenously determined according to fertility choices of cohorts in the fertility stage. The age distribution of this economy is thus determined by (1) households' fertility choices, which are endogenous, and (2) mortality risks after retirement, which are exogenous. Let $\zeta_{j,j+1}$ denote the probability of surviving at age $j+1$ conditional on surviving at age j for each $j \in \{J_R, ..., J\}$ with $\zeta_{J,J+1} = 0$.

Intergenerational Linkages and Initial Endowments: After graduating from high school, agents are endowed with a triple (a_{IVT}, h, ϕ): (1) assets transferred from their parents (a_{IVT}), (2) innate ability (h), which governs education returns in future earnings, and (3) psychic costs of college education (ϕ). They draw the innate ability from a distribution $g^h_{h_p}$, varying with the parents' innate ability level h_p. They also draw the psychic costs from a distribution g^{ϕ}_{h,e_p}, varying with the student's innate ability h and parent's education e_p.

Preferences: Every period, they draw utility from consumption c and leisure l according to a utility function $u(c, l)$. They discount future utility by a factor β. At age J_F, they choose the number of children they have, denoted by $n \in \{0, 1, 2, 3, 4\}$. They draw additional utility from having children in several ways. First, they draw utility from expenditures q per child until the children graduate from high school, captured by a utility function $v(q)$. Note that this expenditure does not affect the child's state variables that are taken into account when education choices are made. Therefore, the function $v(q)$ can essentially be interpreted as another type of consumption good that parents enjoy. The utility $v(q)$ is discounted by a function $b(n)$, increasing and concave in the number of children n. Therefore, having more children (i.e., a higher n) also increases utility, holding other factors constant. Further, based on altruistic motives, parents draw utility by transferring assets to their children after the children graduate from high school. Specifically, the children's expected lifetime utility in that stage, V_{g0}, which depends on the parental transfer, enters the parents' utility. The weight placed on children's utility is given by $\lambda_a \cdot b(n)$, where λ_a denotes the altruistic discount factor.

Costs of Children: Having children entails both monetary and time costs. First, raising children entails a monetary cost of $n \cdot q$, where n and q represent the number of children and the per-child expenditure, both chosen by the household. Second, parents make an inter vivos transfer of $n \cdot a_{IVT}$ when their children graduate from high school, where a_{IVT} is also chosen by the household.[14] In addition to the monetary costs, having a child entails an exogenous time cost of κ units until the child graduates from high school and becomes independent.

Labor Earnings: Before retirement, households choose their working hours to earn labor income. The labor earnings of a household are given by $w_e \eta_{j,e,h,z}(T - l)$; it is a

[14] Children within a household are homogeneous, as assumed in the literature (e.g., Abbott, Gallipoli, Meghir, and Violante, 2019). Thus, education spending and IVT do not vary among children within the household.

product of the market wage (w_e), labor productivity ($\eta_{j,e,h,z}$), and hours worked ($T - l$). The market wage, w_e, depends on the education level $e \in \{HS, CL\}$, where $e = HS$ and $e = CL$ represent high school and college graduates. Labor productivity, $\eta_{j,e,h,z}$, depends on age j, education level e, innate ability h, and idiosyncratic component z. The idiosyncratic component z follows a Markov process $\pi_z(z, z^-)$, where z^- denotes the component in the previous period. T denotes the disposable time that can be devoted to work or leisure where $T = 1 - \kappa \cdot n$ if they have n children who have not graduated high school and $T = 1$ otherwise. The working hours are therefore given by the difference between the total disposable time and the time spent on leisure, $T - l$.

Financial Markets: Financial markets are incomplete due to the lack of state-contingent claims. Households can self-insure against negative labor productivity shocks by savings, accruing interest payments at a rate of r. Households in the working stages can borrow at rate $r^- = r + \iota$ where $\iota > 0$ (i.e., borrowers incur the overseeing costs ι) up to a borrowing limit $\underline{A}$. In contrast, retired households are not allowed to borrow, which is a common assumption in the literature. In addition, college students have access to government-subsidized student loans that carry an interest rate of $r^s = r + \iota_s$. The borrowing limit for these loans is denoted by $\underline{A}_s$.

Firm and Production Technology: A representative firm hires labor and capital in competitive factor markets to produce the final good, Y. The firm operates a constant-returns-to-scale production function, $F(K, L_{HS}, L_{CL})$, using aggregate capital K, efficiency units of labor supplied by high-school graduates (unskilled workers), L_{HS}, and by college graduates (skilled workers), L_{CL}. The firm takes factor prices as given, with the price of the final good normalized to one. Capital depreciates at rate δ, and the firm incurs the associated depreciation cost. Given prices and technology, the firm maximizes its profit by choosing capital and labor inputs:

$$\max_{K, L_{HS}, L_{CL}} F(K, L_{HS}, L_{CL}) - (r + \delta)K - w_{HS}L_{HS} - w_{CL}L_{CL}.$$

Government: The government raises the revenue by levying three types of taxes: consumption, labor income, and capital income taxes, where each tax rate is represented as τ_c, τ_w, and τ_a. In addition, the government collects accidental bequests and devotes them to cover expenditures. They use this revenue to fund (1) the public pension benefits, which gives p units of money to retired households each period, (2) subsidized loans for college students, (3) grants for eligible college students, where the payment per eligible student is represented by $\bar{g}(I)$ where I represents household income when the student is in the education stage (i.e., their parent's age is J_{IVT}), (4) other public expenditures on college education, which capture the public resources required to provide higher education for each student but do not directly enter the household's decision problem, (5) lump-sum transfers ψ that is introduced for replicating the progressivity of labor income tax schedule in a simple way following the literature (e.g., Abbott et al., 2019), (6) cash benefits for households with children under 20 with per-child payment of B, and (7) other expenditures S. The government budget constraint is given as follows:

$$\tau_c \cdot C + \tau_w \cdot (w_{HS}L_{HS} + w_{CL}L_{CL}) + \tau_a \cdot r \int_{\boldsymbol{x}} \mathbb{I}_{a(\boldsymbol{x})>0} a(\boldsymbol{x}) dF(\boldsymbol{x}) + Q$$
$$= p \cdot \mu_{old} + \psi + B \cdot \mu_{j\leq 20} + S \underbrace{+(\iota - \iota_s) \cdot K_s + G + \mu_s \cdot A_s}_{\text{expenditures on college education}}, \tag{4}$$

where C, Q, μ_{old}, $\mu_{j\leq 20}$, K_s, G, μ_s, and A_s represent the total consumption (C), total accidental bequests (Q), population mass of retired households (μ_{old}), that of children under age 20 ($\mu_{j\leq 20}$), the total amount of borrowing by college students (K_s), the total grant payments (G), the population mass of college students (μ_s), and other public expenditures on college education per student (A_s). The term $\tau_a \cdot r \int_{\boldsymbol{x}} \mathbb{I}_{a(\boldsymbol{x})>0} a(\boldsymbol{x}) dF(\boldsymbol{x})$ represents revenue from capital income taxation, where $\boldsymbol{x}$, $a(\boldsymbol{x})$, $\mathbb{I}_{a(\boldsymbol{x})>0}$, and $F(\boldsymbol{x})$ represent the household state vector, household's policy function for savings, an indicator function returning one if $a(\boldsymbol{x})$ is positive, and distribution over the state space.

4.3 Household Problems

Building on the primitives described so far, this section formulates the household problem. To do so, I divide the lifecycle into several stages, summarized in Table 2. The first stage is the education stage, in which households choose whether to attend college or enter the labor market after graduating from high school.

Table 2: The lifecycle stages and the corresponding biological age

Stages	**Corresponding biological age**	
	High-School Graduates	College Graduates
(1) Education stage	20	20–23
(2) Early working stage (before fertility stage)	20–31	24–31
(3) Fertility stage	32	32
(4) Middle working stage (with children)	32–51	32–51
(5) Inter vivos transfer stage	52	52
(6) Late working stage (after children leave home)	52–63	52–63
(7) Retirement stage	64–	64–

(1) Education Stage: At the beginning of the stage, each individual draws their innate ability and school taste from the distributions $g^h_{h_p}$ and g^{ϕ}_{h,e_p}, respectively. In addition, they receive assets from their parents, and parental income determines their eligibility for the grant program. The state variables in this stage therefore consist of parental transfers a_{IVT}, psychic costs ϕ, innate ability h, and parental income I. They compare the expected value for entering the labor market as a high school graduate, $\mathbb{E}V^w$, with the value for enrolling in college V_{g1}, which is the expected lifetime utility conditional on enrolling in college, net of the psychic cost ϕ. They choose college enrollment if the latter is greater than the former; otherwise, they enter the labor market. The decision problem for college enrollment, along with the corresponding value function V_{g0}, is formulated as follows:

$$V_{g0}(a_{IVT}, \phi, h, I) = \max_{e\in\{HS,CL\}} \Big\{(1-\mathbb{I}_{e=CL})\cdot \underbrace{\mathbb{E}_{z_0}[V^w(a_{IVT}, J_E, z_0; HS, h)]}_{\text{Lifetime utility for high school graduates}} + \mathbb{I}_{e=CL}\cdot \underbrace{[V_{g1}(a_{IVT}; h, I) - \phi]}_{\text{Value of attending college}}\Big\}, \quad (5)$$

where $e \in \{HS, CL\}$ indicates the education choice: $e = CL$ means college enrollment and $e = HS$ does entering the labor market as a high school graduate. The indicator $\mathbb{I}_{e=CL}$ takes the value 1 if $e = CL$ and 0 otherwise. $V^w(a, J, z; e, h)$ denotes the value function for workers with state vector (a, J, z, e, h), and thus $\mathbb{E}_{z_0}V^w(a, J_E, z_0; HS, h)$ represents the expected lifetime utility of entering the labor market immediately after high school graduation. The formulation of V^w reduces to a standard consumption–saving problem with endogenous labor supply. The formal representation is provided in Online Appendix E.1. The initial draw of z, z_0, is uncertain and is according to the invariant distribution of z.

V_{g1} in the value function (5) represents the gross value of attending college, not accounting for the psychic costs ϕ. Proceeding to college will increase their lifetime earnings, while this return from education depends on innate ability. In addition to the psychic costs, there are other two types of costs of proceeding to college. First, they have to pay tuition fees p_{CL} while enrolling in college. Second, although they can supply labor and earn some income, they have to spend $\bar{t}$ fraction of their time on study, which is the opportunity costs of going to college. They can fund their consumption and tuition fees through (1) transfers from their parents a_{IVT}, (2) labor earnings by themselves, where the hourly wage is given by the market wage for high-school graduates, w_{HS}, (3) borrowing through student loans, (4) government-provided grants if eligible. The problem for college students and the associated value of attending college V_{g1} are formulated as follows:

$$V_{g1}(a_{IVT}; h, I) = \max_{c,l,a'} \{u(c, l) + \beta\, \mathbb{E}_{z_0}[V^w(a_s(a'), J_E + 1, z_0; CL, h)]\}$$

$$\text{s.t.} \quad (1+\tau_c)c + p_{CL} + a' = a_{IVT} + \bar{g}(I) + (1-\tau_w)w_{HS}(1-\bar{t}-l) + \psi,$$
$$c > 0, \quad l \in [0,\, 1-\bar{t}], \quad a' \geq -\underline{A}_s.$$

Recall that ψ denotes the lump-sum transfer and $\bar{g}(I)$ denotes the grant (cash transfer) provided by the government. Following the literature, I map student loan balances into equivalent asset holdings using the function $a_s(a')$, assuming that fixed payments are made for 20 years (five model periods) following college graduation.[15]

(2) Early Working Stage: The household problem in this period reduces to a standard lifecycle problem in which households choose consumption-saving and leisure-labor decisions. The value function is formulated in Online Appendix E.1.

[15]More specifically, the function is given by $a_s(a') = a' \times \frac{r^s}{1-(1+r^s)^{-5}} \times \frac{1-(1+r^-)^{-5}}{r^-}$.

(3) Fertility Stage: When they reach the age of J_F, they choose how many children they have. The state variables in this stage consist of asset position a, productivity z, education e, and innate ability h. The problem is to choose the number of children n that maximizes expected utility from this point onward, with the value function formulated as follows:

$$V^f(a, z, e, h) = \max_{n \in \{0,1,2,3,4\}} \left\{ V^{wf}(a, J_F, z; e, h, n) \right\},$$

where $V^{wf}(a, j, z; e, h, n)$ in the parenthesis represents the value function for the middle working stage with n children represented below.

(4) Middle Working Stage (with children if applicable): The state variables in this stage consist of asset position a, age j, productivity z, education e, innate ability h, and the number of children n. The value function V^{wf}, after fertility choices until their children become independent (i.e., for age $j \in \{J_F, ..., J_{IVT}-1\}$), is formulated as follows:

$$V^{wf}(a, j, z; e, h, n) = \max_{c,l,q,a'} \{u(c/\Lambda(n), l) + b(n) \cdot v(q) + \beta \mathcal{V}'\},$$

$$\mathcal{V}' = \begin{cases} \mathbb{E}_{z'} \left[V^{wf}(a', j+1, z'; e, h, n) \right] & \text{if } j \in \{J_F, ..., J_{IVT}-2\} \\ \mathbb{E}_{z', \phi_k, h_k} \left[V^{IVT}(a', z'; \phi_k, h_k, e, h, n) \right] & \text{if } j = J_{IVT}-1 \end{cases}$$

s.t.

$$(1+\tau_c)(c+nq) + a' = (1-\tau_w) w_e \eta_{j,e,h,z}(1-l-\kappa n) + nB + \psi + \tilde{a}(a),$$

$$\tilde{a}(a) = \begin{cases} (1+(1-\tau_a)r)a, & \text{if } a \geq 0, \\ (1+r^-)a, & \text{if } a < 0, \end{cases}$$

$$l \in [0, 1-\kappa n], a' \geq -\underline{A}, \;\; z' \sim \pi(z', z), \;\; h_k \sim g_h^{h_k}, \;\; \phi_k \sim g_{e,h_k}^{\phi_k},$$

Here, ϕ_k and h_k denote the psychic costs of college education and innate ability for their children. The household consumption is deflated by the equivalence scale $\Lambda(n)$, depending on the number of children n. The expenditures on children, nq, enter the budget constraint, and utility from children, $b(n) \cdot v(q)$, enters the objective function in the periods with children. V^{IVT} represents the value function for the inter vivos transfer (IVT) stage at $j = J_{IVT}$, which is described in the following paragraph. Households are uncertain about their children's innate ability h_k and psychic costs ϕ_k until the beginning of age J_{IVT}. Note that the expenditures q do not affect children's characteristics (ϕ_k, h_k).[16] The term $\mathcal{R}(a)$ in the budget constraint represents the asset position at beginning of this stage, expressed as a function of the asset position at the end of the previous period, a. If a is negative (i.e., the household borrows), it has to pay the interest rate r^-. If a is positive (i.e., the household saves), it earns the interest rate r net of the capital tax τ_a.

(5) Inter Vivos Transfers Stage: At the beginning of age J_{IVT}, households decide how much assets to transfer to their children. This timing corresponds to when the children graduate from high school and face the college enrollment choices. Two characteristics of their children are realized at this point: their psychic costs ϕ_k and innate

[16] I discuss this assumption in Online Appendix E.3.

abilities h_k. After observing the realized characteristics, households choose the optimal amount of per-child transfer a_{IVT}, formulated as follows:

$$V^{IVT}(a, z; \phi_k, h_k, e, h, n) = \max_{a_{IVT} \geq 0} \Big\{ V^w(a - \tilde{a}_{IVT}, J_{IVT}, z; e, h) + b(n) \cdot \lambda_a \cdot V_{g0}(a_{IVT}, \phi_k, h_k, I) \Big\},$$

where $\tilde{a}_{IVT} = \frac{n \cdot a_{IVT}}{1+(1-\tau_a)r}$ and the parent's state vector pins down household income I. $V^w(a, j, z; e, h)$ represents the value function during the late working stage after children leave the household. It captures the lifetime utility that households obtain from age j onward, conditional on asset holdings a and the other state variables. Driven by altruism, they care about the lifetime utility of each child—$V_{g0}(a_{IVT}, \phi_k, h_k, I)$, as defined in equation (5)—which they discount by an altruistic factor λ_a. Higher inter vivos transfers increase parents' utility through this altruistic channel, but they also reduce the assets available for the remainder of the parents' lifetime. Households therefore choose transfers by trading off these two effects.

(6) Late Working Stage: At this stage for households aged $j \in \{J_{IVT}, ..., J_R - 1\}$, the household problem is similar to the early working stage: they make consumption-saving and leisure-labor decisions until retirement. The value function is formulated in Online Appendix E.1.

(7) Retirement Stage: At the beginning of age J_R, households are forced to retire from the labor market. After that, they spend all their time on leisure and make consumption-saving decisions. Two additional aspects distinguish this stage from earlier choice problems: they receive the pension benefit p from the government and face uncertainty about the next period's survival. The value function is formulated in Online Appendix E.1.

4.4 Stationary Equilibrium

I solve the stationary equilibrium of this economy. In equilibrium, households maximize their expected utility, the firm maximizes its profit, and the government budget is balanced. Factor prices are determined to clear markets. Stationarity implies that the distribution over state variables is invariant. Importantly, the age distribution is determined endogenously according to households' fertility choices. See Online Appendix E.2 for the detailed definition of equilibrium and Online Appendix F for the computational algorithm for solving the equilibrium.

5 Calibration

I now parameterize the model presented in the previous section so that it is suitable for quantifying the roles of income uncertainty and income-tested education subsidies on fertility, as well as their macroeconomic implications. I first explain the calibration strategy in Section 5.1 and discuss the model fit in Section 5.2. Appendix B.3 validates the calibrated model by showing that its implied fertility response to pro-natal cash transfers is broadly consistent with the existing empirical evidence.

5.1 Strategy

The calibration follows a standard two-step procedure. In the first step, I determine a subset of parameters outside the model, either by estimating them directly from the data, assigning them based on observable data moments, or adopting estimates from the empirical literature. In the second step, I jointly determine the remaining parameters by minimizing the distance between the model-implied moments and their empirical counterparts. Let $\boldsymbol{d}$ be a vector of n targeted moments observed in the data, and let $\boldsymbol{m}(\boldsymbol{\theta})$ be the corresponding vector of model-implied moments, where the parameter vector $\boldsymbol{\theta}$ is of dimension $k \leq n$. I determine $\boldsymbol{\theta}$ by minimizing the following objective function:

$$\min_{\boldsymbol{\theta}}[\boldsymbol{d} - \boldsymbol{m}(\boldsymbol{\theta})]'[\boldsymbol{d} - \boldsymbol{m}(\boldsymbol{\theta})]$$

I determine 23 parameters internally, targeting 47 moments (i.e., $n = 47$ and $k = 23$).

To compute the empirical moments in the second step, I use the Japanese Panel Survey of Consumers (JPSC), a panel survey of Japanese women and their household members. It started in 1993 with a representative sample of 1,500 women aged $24 - 34$ and contains information about, for example, their income, educational background, marital status, fertility, and expenditures to detailed categories, including those on children's education. I focus on the cohort born in 1959-69, the oldest cohort of this survey, especially to compute the completed fertility and intergenerational mobility of education. I keep only married households as in previous works (e.g., Daruich and Kozlowski, 2020) because the model focuses on choices made within married households such as fertility and educational investments.[17] Unless otherwise mentioned, targeted moments described below are computed based on the JPSC's 1959-69 cohort data.

In what follows, I discuss the functional forms and explain how each parameter is determined, that is, whether it is determined externally in the first step or internally in the second step. For parameters determined internally, I also discuss which empirical moments are informative for their identification. Since the model features multiple interconnected choice margins, each parameter generally affects several empirical moments simultaneously. Nevertheless, certain moments are particularly informative for identifying specific sets of parameters. Table A.2 summarizes the parameter values determined in the first step, while Table A.3 reports those determined in the second step. Table A.4 and Figures 2–3 summarize the targeted moments in the data and their model counterparts, which are discussed in detail in the next section.

Preferences: For preferences, three functions must be specified: the instantaneous utility from consumption and leisure, $u(c, l)$; the utility from expenditures on children, $v(q)$;

[17]Hence, an agent or household in this model refers to households with two individuals. Then, "children" in this model can also be interpreted as a household unit. That is, having n children can be interpreted as reproducing $n/2$ units of households. For example, if a household has two children, it means reproducing one household unit with two individuals.

and the child discounting function, $b(n)$. I specify these functions as follows:

$$u(c,l) = \frac{(c^{\mu}l^{1-\mu})^{1-\gamma}}{1-\gamma}, \quad v(q) = \lambda_q \frac{q^{1-\gamma_q}}{1-\gamma_q}, \quad \text{and} \quad b(n) = n^{\gamma_n}.$$

The weight parameter on consumption relative to leisure, μ, is internally determined to match the average amount of time spent in market work. The parameters governing utility from expenditures on children, λ_q and γ_q, are determined to match the expenditures on children across income distribution. Given these parameters, the curvature parameter for consumption and leisure, γ, governs the degree of diminishing marginal utility and thereby influences the relationship between income and labor supply. Because μ primarily determines the average level of hours worked, moments capturing variation in working hours across the income distribution help identify γ. The altruistic discount factor λ_a is determined to match the average level of inter vivos transfers (IVTs) for college students.[18] Given the altruistic discount factor, the time discount factor β is calibrated to match the target interest rate (a four-year rate of 20%). Completed fertility across the income distribution and the distribution of the number of children help identify the time cost of children, κ, and the curvature parameter governing child discounting, γ_n.

Financial Markets: The borrowing limits are externally calibrated to $\underline{A} = 20$ million yen and $\underline{A}_s = 2.88$ million yen based on data from the Family Income and Expenditure Survey (FIES) and the Student Life Survey (2018). These values correspond to 3.6 and 0.52 times the average household income at age 28, respectively. The borrowing wedges, ι and ι_s, are determined internally by targeting the share of working households with negative net worth and the average income share of student loans within revenue for college students.

Intergenerational Transmission of Ability and Tastes: The psychic cost of college, ϕ, is given by

$$\phi = \psi_{CL} \exp(-\nu h)\tilde{\phi},$$

where $\tilde{\phi}$ captures the stochastic component of psychic costs, the distribution of which depends on parental education e_p. Following Daruich and Kozlowski (2020), $\tilde{\phi}$ is distributed over the interval $[0,1]$ with cumulative distribution function

$$G_{e_p}(\tilde{\phi}) = \begin{cases} \tilde{\phi}^{\omega} & \text{if } e_p = 0 \text{ (high school)}, \\ 1-(1-\tilde{\phi})^{\omega} & \text{if } e_p = 1 \text{ (college)}. \end{cases}$$

The parameter ω governs the differential intergenerational transmission of tastes for schooling across parental education groups. The variation in college enrollment across parental education is informative for identifying this parameter ω. The parameter ψ_{CL} determines the overall scale of psychic costs and is therefore determined to replicate the average college enrollment rate. The term $\exp(-\nu h)$ allows higher-ability individuals to face lower psychic costs, as is standard in the literature, and the parameter ν governs the

[18] Although some parents whose children do not enroll in college make IVT in the model, the amount is negligibly small. One reason is that the marginal gains from IVT are more significant if their children attend college, as students are financially constrained, primarily because of their limited earning ability.

degree of educational sorting by ability. IVTs across parental income groups help identify the parameter ν, given the intergenerational persistence of innate ability.[19]

The intergenerational transmission of innate ability follows the process

$$\ln(h) = \rho_h \ln(h_p) + \varepsilon_h, \quad \varepsilon_h \sim N(0, \sigma_h^2).$$

where h denotes the innate ability of children and h_p denotes the innate ability of their parents. The parameter ρ_h governs the degree of intergenerational persistence in ability, while σ_h controls the dispersion of idiosyncratic ability shocks. The persistence parameter ρ_h is internally calibrated to replicate the intergenerational income elasticity—that is, the coefficient from a regression of children's log income on parents' log income. The dispersion parameter σ_h is also internally calibrated to match income levels across the income distribution relative to median income.

Income Process: I follow Daruich and Kozlowski (2020) and specify the efficiency labor units of an individual with age j, education e, innate ability h, and idiosyncratic productivity z, denoted by $\eta_{j,e,h,z}$, as follows:

$$\ln \eta_{j,e,h,z} = \ln[f^e(h)] + \gamma_{j,e} + z, \tag{6}$$

$$f^e(h) = h + \mathbb{I}_{e=CL} \cdot (\alpha_{CL} h^{\beta_{CL}}), \tag{7}$$

$$z' = \rho_{z,e} z + \zeta, \qquad \zeta \sim N(0, \sigma_{z,e}^2). \tag{8}$$

The function $f^e(h)$ captures the earnings potential based on innate ability and educational attainment. The parameter α_{CL} in the function $f^e(h)$ is internally determined to match the wage premium of college graduates relative to non-college workers, while the curvature parameter β_{CL} is determined to match the income variance within the college-educated group.

The term $\gamma_{j,e}$ captures the age profile of earnings, which is allowed to differ across education groups, while z represents an idiosyncratic productivity component that evolves stochastically over the lifecycle. I estimate the age profile of expected labor earnings for each education group, $\gamma_{j,e}$, by regressing hourly wages on a second-order polynomial in age using the JPSC data. As reported in Table A.1, the age-earnings profile is steeper for college graduates than for non-college workers, although the difference is modest relative to the U.S. evidence (e.g., Abbott et al., 2019).

To estimate the first-order autoregressive process for the stochastic income component z separately for each education group, I construct a four-year series of household wage measures for each couple and estimate the residual wage process. The estimated persistence parameters, $\rho_{z,e}$, are 0.938 for high school graduates and 0.965 for college graduates,

[19] To understand why, note that the amount of IVTs required to induce college attendance depends on both the returns to education and the psychic costs of college. Since the returns to education are primarily disciplined by other moments discussed below, variation in psychic costs is particularly informative about variation in IVTs. In the model, higher-ability children face lower psychic costs of college attendance, implying that, all else equal, households with lower-ability children must provide larger IVTs to incentivize college enrollment. Because children's innate ability is correlated with parental ability, and parental ability is a key determinant of the income distribution, the distribution of IVTs across parental income groups is informative for identifying ν.

while the corresponding income volatility parameters, $\sigma_{z,e}$, are 0.036 and 0.050, respectively.[20]

These estimates are broadly consistent with previous studies using U.S. data that estimate income processes based on household-level measures over multi-year intervals. The closest comparison is Daruich (2026), who uses the NLSY data to estimate the process separately for college and non-college households using four-year intervals. The study estimates a persistence parameter of 0.916 for non-college graduates and 0.966 for college graduates, which are close to my estimates based on Japanese data (0.938 for non-college and 0.965 for college graduates). Their estimated innovation standard deviations are 0.031 for non-college graduates and 0.046 for college graduates, also close to my estimates of 0.036 and 0.050, respectively. While Krueger et al. (2025) do not estimate the income process separately by education group, their estimates of the persistent income shocks based on the PSID are also broadly similar, implying a four-year persistence parameter of 0.950 and an innovation standard deviation of 0.033 (a variance of 0.011).

Production Technology: The aggregate production function $F(K, L_{HS}, L_{CL})$ is specified as follows:

$$F(K, L_{HS}, L_{CL}) = ZK^{\alpha}L^{1-\alpha}, \text{ where } L = \left[(1-\omega_{CL})(L_{HS})^{\chi} + \omega_{CL}(L_{CL})^{\chi}\right]^{1/\chi}.$$

Here, Z denotes factor-neutral productivity, and $\omega_{CL} \in (0, 1)$ governs the relative productivity weight of college-graduate workers. The parameter χ determines the elasticity of substitution between college- and non-college-graduate labor. I set $\chi = 0.7$ based on Kawaguchi and Mori (2016), who estimate the elasticity of substitution between college and non-college workers using variation in the skill premium and relative labor supply observed in the Japanese Labor Force Survey over the period 1986–2008. I set the capital share $\alpha = 0.31$ based on JIP database (2008) by the Research Institute of Economy, Trade and Industry (RIETI) in Japan. I determine the three parameters (δ, ω_{CL}, and Z) internally to normalize equilibrium prices, targeting a four-year interest rate of 20% and unit marginal products for both skilled and unskilled labor.

Government: Tax rates are set to $\tau_c = 0.1$, $\tau_w = 0.33$, and $\tau_a = 0.50$ in the benchmark. The consumption tax rate of 10% reflects the current consumption tax rate in Japan, whereas the average tax rates for labor income and capital income, τ_w and τ_a, are taken from the estimates by Gunji and Miyazaki (2011), who estimate the average labor income tax (including social security premiums) and capital income tax using Japanese administrative tax statistics. The lump-sum transfer, ψ, is internally determined to match the ratio of the variance of log net income to that of log gross income. The pension benefit p is externally set so that the government provides ¥70,000 per person per month (i.e., $p = ¥70,000 \times 2$ persons), consistent with the amount reported by Japan Pension Service (JPS).[21] The monthly payment of ¥70,000 for two persons amount to 31% of the average household monthly income at age 28. The cash transfer, B, is given as ¥10,000 per child

[20] Details of the estimation procedure are provided in Appendix B.

[21] See, https://www.nenkin.go.jp/international/japanese-system/nationalpension/nationalpension.html.

per month, approximating the actual payment.[22] The monthly payment of ¥10,000 corresponds to 2% of the average monthly household income at age 28. I internally calibrate A_s to match government expenditure on college education as a share of GDP, which is 0.3%, in the benchmark model.[23] This corresponds to about 1.8% of total government expenditure in both the model and the data, a small amount that is consistent with the fact that the Japanese government provides relatively little subsidy to college education compared with other countries (Online Appendix D). The other expenditure S is set so that the government budget constraint is balanced in the benchmark and fixed throughout the counterfactual experiments. In the baseline model, I set $g(I) = 0$ for all I, meaning that college grants are not available. This specification reflects the period before the introduction of the grant program in 2020.

Miscellaneous: The survival probability $\{\zeta_{j,j+1}\}_j$ is externally set based on the Vital Statistics (2019).[24] The time devoted to studying, $\bar{t}$, is also externally determined and set to 0.21, given that college students spend 21% of their time on college-related activities, such as attending classes and studying (Student Life Survey, 2018). The pecuniary cost of college education, p_{CL}, is determined internally in the second step. College enrollment rates and IVTs across the parental income distribution jointly help identify this parameter. I adopt the OECD modified equivalence scale, defined as $\Lambda(n) = 1.5 + 0.3 \times n$.

5.2 Model Fit

The calibrated model fits the targeted moments well, including key cross-sectional moments related to college enrollment, fertility, parental transfers, and labor supply. Table A.4 summarizes the targeted moments in the data and the model, while Figures 2–4 illustrate the model fit for the key targeted moments across disaggregated groups. In what follows, I discuss the model fit with respect to the key moments.

First, Figure 2 (a) compares labor income across income quintiles in the data and the model and shows that the model matches the data closely. Part of the observed income differences reflects variation in labor supply. Figure 2 (b) shows that higher-income households work longer hours on average, a pattern that is also well captured by the model.

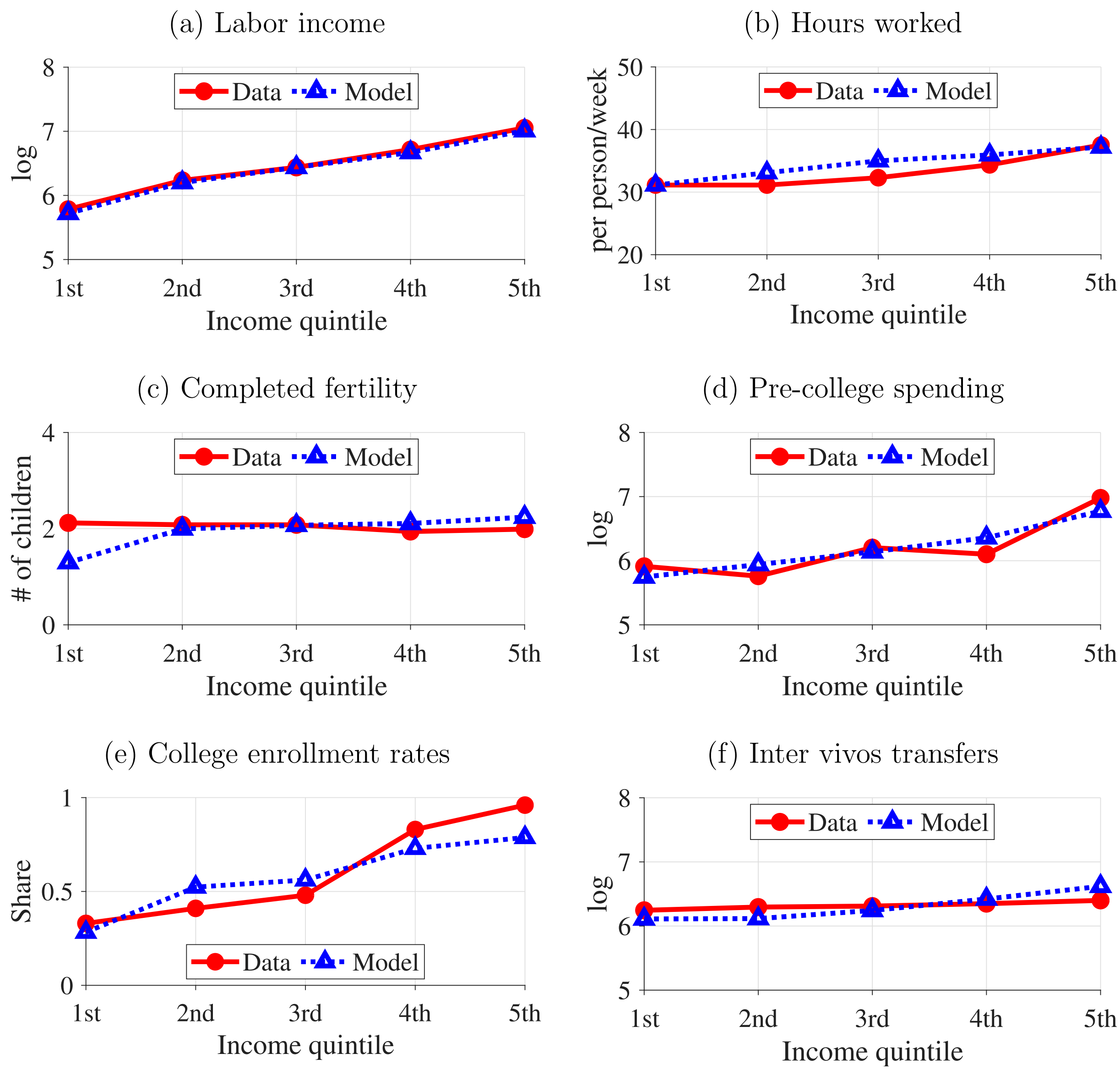
Figure 2 (c) shows completed fertility by parental income quintile. In the data, completed fertility is approximately flat across the income distribution at around two children per household. The model reproduces the overall flat profile observed in the data, although it understates fertility among households in the lowest income quintile. Figure 3 provides an additional perspective on fertility by comparing the distribution of the number of children in the data and the model. In our sample of married couples, roughly half of married couples have two children, while the remainder is approximately evenly split between one and three children. Figure 3 shows that the model replicates this pattern.

[22]See, for example, the website by the Ministry of Health, Labour and Welfare (MHLW), https://www.mhlw.go.jp/english/topics/child-support/.

[23]For data on public expenditures on tertiary education, see https://www.mext.go.jp/b_menu/shingi/chukyo/chukyo0/toushin/attach/1335651.htm (in Japanese).

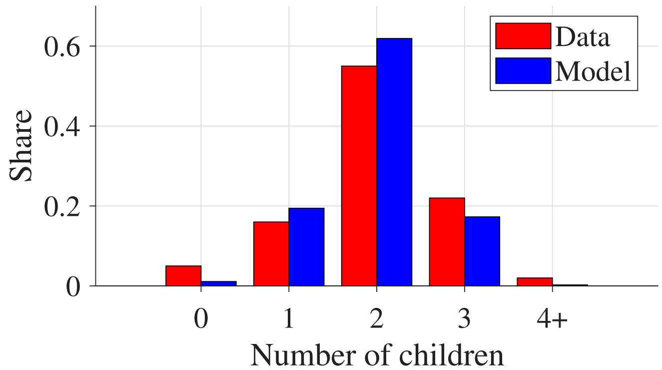
[24]See, https://www.mhlw.go.jp/english/database/db-hw/outline/index.html.

Figure 2: Moments across income quintiles: data and model



Note: In the model, income quintiles are assigned based on household income at age J_F, when households make their fertility choices. The data are drawn from the JPSC (1993–2019). The sample consists of married couples in which the wife was born between 1959 and 1969. Income is defined as the sum of the husband's and wife's labor income and is measured as the average over ages 32–35, corresponding to its model counterpart. Hours worked are reported on a per-person basis and are computed by dividing the total hours worked by the husband and wife by two. Pre-college educational spending refers to total educational expenditures incurred before college enrollment, including school tuition, private tutoring, and extracurricular activities. Inter vivos transfers refer to the amount of money parents send to children enrolled in college.

Figure 3: Distribution of number of children

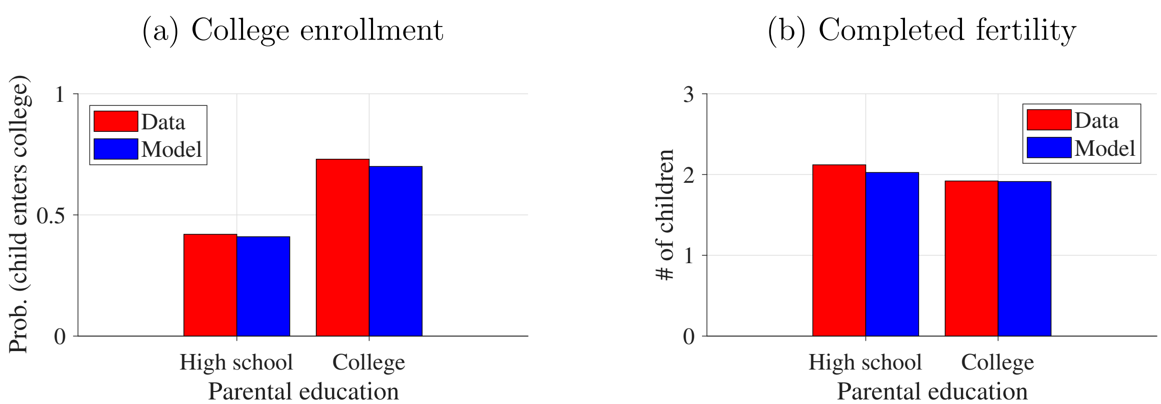


Note: The data are drawn from the JPSC (1993–2019). The sample consists of married couples in which the wife was born between 1959 and 1969. The number of children corresponds to completed fertility.

Figure 2 (d) presents pre-college educational spending by parental income quintile. The data exhibits a strong positive income gradient, and the model matches this pattern closely across the entire income distribution. Figure 2 (e) shows college enrollment rates by parental income quintile. The model captures the positive relationship between parental income and college attendance, whereby children from higher-income households are more likely to enroll in college.

Finally, Figure 2 (f) reports parental transfers to children enrolled in college. In the data, transfers are relatively flat across parental income quintiles. This pattern is consistent with the fact that parents bear the primary responsibility for financing higher education in Japan: conditional on a child attending college, parents provide substantial financial support regardless of income. The model reproduces the overall flatness of transfers conditional on college enrollment, although the implied income gradient is somewhat steeper than in the data.

Figure 4: Moments across parental education



Note: The data are drawn from the JPSC (1993–2019). The sample consists of married couples in which the wife was born between 1959 and 1969. College graduates are defined as households in which either the husband or the wife has a college degree, while all other households are classified as high-school graduates.

The model also replicates key moments across parental education, particularly those related to fertility and college enrollment. Figure 4 (a) shows college enrollment rates by parental education. In the data, while the enrollment rate is 0.43 among children of high-school graduates, it rises to 0.72 among children of college graduates. As the figure shows, the model closely matches these enrollment rates and the substantial education gradient in college attendance.

Figure 4 (b) presents completed fertility by parental education. College graduates have lower completed fertility (1.92) than high-school graduates (2.12). In the calibration, I target the fertility ratio between the two groups (1.92/2.12 = 0.91), while the overall fertility level is disciplined by targeting fertility across income quintiles. The model generates a fertility ratio of 0.94, which is close to the target, and consequently replicates the fertility differential by parental education observed in the data.

6 Experiments

In this section, I use the calibrated model to address the following questions: (1) What are the quantitative impacts of income uncertainty on fertility? (2) What are the effects of income-tested college subsidies on fertility, and how large is the insurance effect? (3) What are the macroeconomic implications of the subsidies? The first two questions serve as the quantitative investigation of the two propositions presented in Section 3: one establishing a negative relationship between fertility and income risk, and the other showing an insurance effect of income-tested subsidies on fertility.

6.1 Roles of Income Uncertainty on Fertility

To examine the quantitative effects of income uncertainty on fertility, I eliminate income risk from the model. Specifically, I set the stochastic component, z, in the wage equation (6) equal to zero, eliminating idiosyncratic wage fluctuations so that the wage process consists only of the deterministic component:

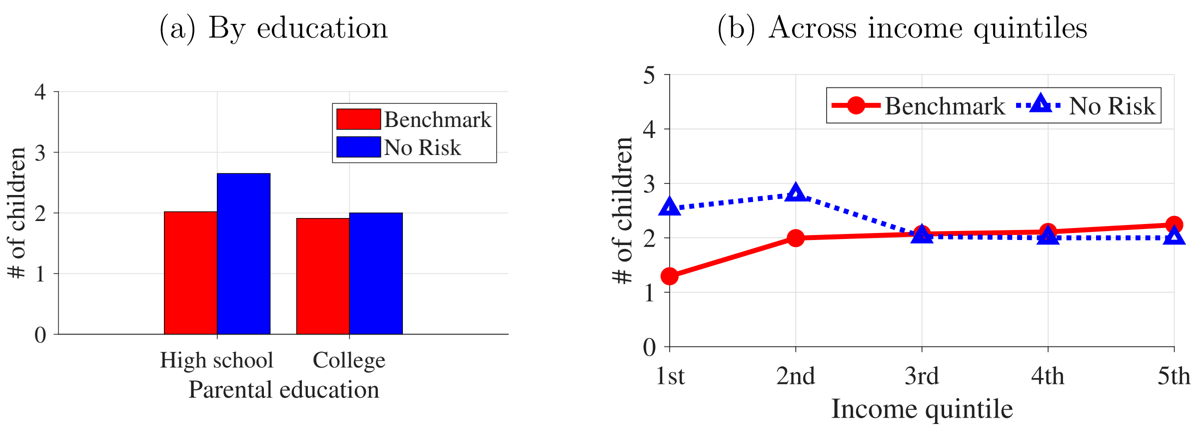
$$\ln \eta_{j,e,h,z} = \ln[f^e(h)] + \gamma_{j,e}. \tag{9}$$

This specification is equivalent to assuming that all households start with the mean value of the stochastic component ($z_0 = 0$) and that the stochastic component has no volatility throughout the lifecycle ($\sigma_{z,e} = 0$ for each e). To isolate the pure effects of income uncertainty on fertility, I conduct this exercise under partial equilibrium, holding prices and tax rates fixed at their baseline levels.

Eliminating income risk increases completed fertility by 15.8% on average. To put its magnitude into perspective, applying a 15.8% increase to Japan's 2024 total fertility rate of 1.14—which was the sixth lowest among the 38 OECD countries—yields a fertility rate of approximately 1.32. This level is substantially closer to the OECD average of 1.40. Put differently, the implied effect is large enough to move Japan from having the sixth-lowest fertility rate in the OECD to a level close to the OECD average.

Figure 5 presents completed fertility across parental education groups and income quintiles in both the baseline economy and the counterfactual economy without income risk. The figure shows that the fertility effects are concentrated among less-educated and lower-income households. For example, eliminating income risk leads to a 31.1% increase in completed fertility among high school graduates, compared with only a 4.7% increase among college graduates. This result is intuitive, as less-educated households have a more limited ability to self-insure against income risk because of their lower earnings potential. The same logic applies to fertility responses across the income distribution, with the effects concentrated among households in the lower income quintiles. Although fertility declines slightly in the highest income quintile, it is important to note that individuals are not assigned to the same income quintiles in the benchmark and counterfactual economies. This is because the wage process in the counterfactual economy excludes the stochastic component, thereby altering the income distribution and households' positions within it. As a result, the comparison does not track the same set of individuals across economies within a given income quintile. Therefore, these small changes are likely to reflect compositional differences across income quintiles.

Figure 5: Fertility outcomes with and without income risk



Note: In the no-risk economy, the stochastic component of income is eliminated by assuming that all households begin life with the expected income realization ($z_0 = 0$) and by setting the innovation variance ($\sigma_{z,e}$) to zero, so that the wage process depends only on the deterministic components. Income quintiles are assigned based on household income at age J_F, when households make their fertility choices. Fertility outcomes are computed in partial equilibrium, holding prices and tax rates fixed at their baseline values.

6.2 Roles of an Income-tested College Subsidy

The counterfactual experiment in the previous section demonstrates that eliminating income risk significantly increases fertility, particularly among households at the lower end of the income distribution. I now turn to the policy implications and examine the effects of income-tested college subsidies on fertility (Section 6.2.1), as well as their broader macroeconomic implications (Section 6.2.2). To do so, I introduce into the calibrated model an income-tested college subsidy that approximates the program implemented in Japan in recent years. More specifically, I allow for an income-tested subsidy for students in low-income households, whose income is below $\bar{I}$, and students in these households receive an amount g each period while they are in college. The grant function $\bar{g}(I)$ is formulated as follows:

$$\bar{g}(I) = \begin{cases} g & \text{if } I < \bar{I} \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

The income threshold $\bar{I}$ in the actual policy approximately corresponds to the 40th percentile of the household income distribution, and the payment amounts to approximately two-thirds of the average college student's expenses, including both tuition fees and living expenses.[25] I set $\bar{I}$ and g based on this information, considering the income distribution and students' expenditures in the benchmark model (initial steady state).

In the computation, I refer earnings potential ($w_e \eta_{j,e,h,z}$) instead of actual income (earnings potential multiplied by hours worked) to determine eligibility and the threshold for the subsidy $\bar{I}$. This approach rules out the possibility of households engaging in strategic behavior by reducing their labor supply to qualify for the subsidy.[26]

6.2.1 Effects on Fertility

I simulate the introduction of the subsidy in a partial equilibrium setup, in which prices, tax rates, and distributions are fixed as in the benchmark, and examine its direct effect on fertility. I then decompose the direct effect into two components. The first is the insurance effect, which was discussed in Section 3. Although the analytical model considers an income-tested subsidy with an expected value of zero, the actual policy considered here has a positive expected value. As a result, an additional effect arises: the price effect. This captures the reduction in the expected monetary cost of children's education, abstracting from the insurance value of the subsidy that becomes available when households experience bad income realizations. The insurance effect, in turn, captures the fertility response arising from the insurance value of state-contingent educational support, holding expected education costs constant.

I decompose the direct effects into insurance and price effects as follows. First, I compute fertility responses under a counterfactual in which the expected value of the college grant, conditional on parental income at the fertility stage, is paid with certainty if a child enters college. For example, households in the top income quintile at the fertility stage are unlikely, if not impossible, to be eligible for the grant in the future when their children attend college. As a result, the expected value of the grant is low for these households. In contrast, households in the bottom income quintile are much more likely to be eligible if their children attend college, implying a higher expected grant value. The resulting fertility response corresponds to the price effect of the subsidy, as it captures the impact of the subsidy on fertility through a reduction in the expected education costs. The insurance effect is then obtained as the difference between the overall direct effect of the subsidy and the price effect, capturing the insurance provided by the subsidy when households experience bad income realizations.[27]

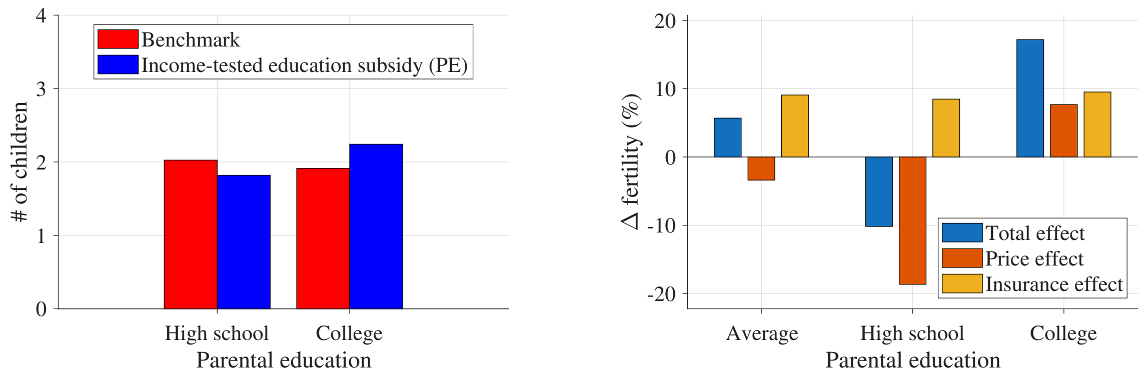
The main results of the exercises are as follows. First, while the introduction of the income-tested college subsidy increases aggregate fertility by 5.7%, this aggregate effect

[25] While the policy initially targeted students from low-income households when it was introduced in 2020, its coverage has gradually expanded in recent years. Although the income threshold $\bar{I}$ and payment g in the actual system vary with certain characteristics of households and students, such as family structure and whether the student commutes to college from home, I abstract away from those details for simplicity.

[26] Although such strategic behavior can occur in practice and be relevant for policy design, there are ways to mitigate this issue to some extent—for instance, by determining eligibility based on income from prior years. Therefore, for simplicity, I consider a setting that entirely rules out such behavior.

[27] See Appendix A.2 for a formal description of the decomposition procedure.

Figure 6: Effects of the income-tested college subsidy on fertility (by parental education)



(a) Fertility (b) Decomposition results

Note: The income-tested education subsidy provides cash transfers g to college students only if their parents' income falls below a threshold $\bar{I}$, which corresponds to the 40th percentile of the income distribution. The fertility effects of the policy are computed under a partial-equilibrium setup in which prices, tax rates, and the household distribution are held fixed at their benchmark levels. In panel (b), the total effect is decomposed into a price effect and an insurance effect. The price effect is computed as the fertility response under a counterfactual in which the expected value of the college grant, conditional on parental income at the fertility stage, is paid with certainty if a child enrolls in college. The insurance effect is computed as the difference between the total effect and the price effect and thus captures the value of state-contingent educational support while holding expected college costs constant. See the main text and Appendix A.2 for details of the decomposition formula.

masks substantial heterogeneity across education groups. Fertility increases by 17.2% among college-graduate parents but decreases by 10.2% among high-school-graduate parents. Figure 6 (a) shows fertility by parental education in both the benchmark economy and the economy with the subsidy. While less-educated parents have higher fertility in the benchmark, the subsidy reverses this pattern: fertility becomes higher among college graduates than among high-school graduates.

Second, the decomposition exercise shows that these heterogeneous effects are driven primarily by differences in the price effect. As shown in Figure 6 (b), the price effect is positive for college graduates but negative and sizable for high-school graduates. In other words, a reduction in college costs increases fertility among college graduates but decreases fertility among high-school graduates. The intuition is as follows. In the benchmark economy, the probability that children attend college is relatively low among high-school-graduate parents (41%). As college costs decline, more children of high-school graduates choose to attend college. Although college becomes cheaper, attending college remains substantially more costly than not attending college because the grant does not fully cover college costs. As a result, the reduction in college costs increases the expected amount of asset transfers per child and makes childrearing effectively more expensive for these households, reducing fertility through the quantity–quality trade-off. In contrast, children of college-graduate parents already have a high probability of attending college in the benchmark economy (70%). For these households, the subsidy frees up resources that would otherwise be used to finance college expenditures, thereby generating a positive income effect that increases fertility. To summarize, the same subsidy generates qualitatively different price effects because college attendance probabilities differ substantially across education groups in the benchmark economy.

Finally, Figure 6 (b) shows that the insurance effect is positive and sizable for both education groups, increasing fertility by 8.5% among high-school graduates and 9.5% among college graduates. Although the quantitative effects are similar across the two groups, the underlying economic mechanisms are likely different. On the one hand, college graduates may benefit from the insurance component primarily because their children are more likely to possess characteristics associated with college attendance, such as lower disutility from education or higher learning ability. As a result, a negative income shock may place these households in a particularly costly situation, preventing them from providing financial support for their children's college education despite its high return. The income-tested college subsidy partially insures against such states, thereby increasing fertility. On the other hand, high-school graduates may benefit from the insurance component primarily because their lower lifetime earnings limit their ability to self-insure against bad income shocks. As Figure 6 (b) shows, the insurance effect plays a central role in generating the positive aggregate fertility response. The price effect alone reduces aggregate fertility because of the quantity–quality trade-off faced by less educated households, but the insurance effect is sufficiently strong to more than offset this force, resulting in a positive overall impact on fertility.[28]

Applicability Beyond Japan: While the central mechanism—that income uncertainty reduces fertility, and policies that insure against income risk can increase fertility—likely applies beyond Japan, the effective implementation of such policies may differ across contexts. Consider an extreme case in which, unlike in Japan, parents do not contribute to college costs at all and students fully finance their education through borrowing, either because of cultural norms regarding parental responsibility or institutional features such as the availability of student loan programs. In such an environment, parents do not directly face the financial burden of future education costs when making fertility decisions. However, if parents themselves have outstanding student loans, their loan repayments reduce disposable income during the child-rearing years. The resulting reduction in disposable income increases the risk of being unable to cover child-related expenditures or maintain one's own consumption, thereby discouraging fertility. This interaction between disposable income, risk, and fertility is consistent with the quantitative results presented above: fertility responses to income uncertainty are strongest among lower-income households, whose ability to self-insure against income shocks is more limited. In such settings, the relevant concern shifts from financing children's college attendance to repaying one's own student debt while raising children. Income-contingent loan programs—under which repayment obligations adjust to realized income—may then operate in a manner analogous to the income-tested subsidies considered here. By reducing the burden of student loan repayments in bad income states, such programs provide insurance against costly states in which households face both child-related expenditures and limited financial resources.

6.2.2 Macroeconomic Implications

I now examine the macroeconomic implications of the policy by solving the model in general equilibrium and allowing factor prices, tax rates, and the household distribution

[28] A similar pattern emerges across income quintiles; see Figure G.7.

to adjust. I first discuss the long-run effects, then consider an expansion of the policy, and finally analyze the transitional dynamics.

Long-run effects: The main result here is that the fertility responses amplify a number of the policy's key outcomes, such as college enrollment and intergenerational mobility. I highlight the roles of endogenous fertility by solving the *exogenous fertility* version of the model, in which the household decision rule (policy function) for fertility, $\bar{n}(a, z, e, h)$, are fixed as in the benchmark. Note that this exogenous fertility setup is common in the macroeconomic literature on college subsidies (e.g., Krueger and Ludwig, 2016; Matsuda and Mazur, 2022). Comparing the results under exogenous fertility with those under endogenous fertility reveals the role of fertility choices in evaluating the macroeconomic implications of a college subsidy. Table 3 reports the main results under both endogenous and exogenous fertility.

Table 3: Long-run macroeconomic implications (endogenous vs. exogenous fertility)

	Fertility setup	
	Endogenous	Exogenous
Education		
College enrollment (Δ p.p.)	+6.2	+5.9
Enrollment rate by parents' education (Δ p.p.)		
High-school graduates	+9.4	+6.4
College graduates	+1.4	+0.7
Inequality and mobility		
Average wage ratio (college/high school; Δ %)	+6.1	−4.6
Intergenerational persistence of income (Δ %)	−9.2	−8.5
Macroeconomic aggregates		
Output per capita (Δ %)	+1.3	+4.0
Labor income tax (Δ p.p.)	+0.11	−0.47
Welfare		
Consumption equivalence (%)	+6.4	+3.6

Note: In the exogenous fertility setup, the household fertility decision rule (policy function), $\bar{n}(a, z, e, h)$, is fixed at its benchmark value. Welfare gains are reported as consumption-equivalent variations for newborn agents behind the veil of ignorance.

The main findings are summarized as follows. First, the policy introduction results in a 6.2 p.p. increase in the college enrollment rate under endogenous fertility, which is slightly higher than a 5.9 p.p. increase under exogenous fertility. Under both exogenous and endogenous fertility, the grants enable some children who would otherwise be unable to afford college to enroll. Also, an increase in the college enrollment rate is amplified in the long run through a distributional effect: the composition of college graduates in younger generations increases, and when they reach parenthood, their children are more likely to enroll in college due to intergenerational linkages via assets, abilities, and tastes.[29]

[29] A detailed decomposition exercise can be found in Online Appendix G.2.

Under endogenous fertility, the college enrollment rate can increase through an additional channel. As discussed in Section 6.2.1, the subsidy raises fertility particularly among college graduates, whose children are more likely to attend college because of the intergenerational transmission of abilities and preferences. When these children become parents themselves, their own children are also more likely to inherit traits that favor college attendance. As a result, the subsidy generates a gradual increase in college enrollment across generations.

A higher college enrollment rate increases the supply of college graduates in the labor market, putting downward pressure on their relative market wage. The ratio of average hourly wages between college and high-school graduates, however, does not necessarily decline because hourly wages reflect not only market wages but also the average labor productivity of each education group. While the wage ratio falls under exogenous fertility, consistent with previous studies that abstract from endogenous fertility (e.g., Krueger and Ludwig, 2016), it increases under endogenous fertility. This result is driven by a compositional effect. Under endogenous fertility, highly educated and high-ability parents are more likely to have children because they benefit more from the insurance provided by the income-tested subsidy, given that their children are more likely to possess traits associated with college attendance. As a result, the additional children born because of the policy are, on average, more likely to attend college and to have high labor productivity. This raises the average productivity of college graduates and more than offsets the downward pressure on their relative market wage.

Intergenerational persistence of income declines more substantially under endogenous fertility. This finding is consistent with studies that emphasize the role of fertility differentials in shaping intergenerational mobility (Daruich and Kozlowski, 2020; Seshadri and Zhou, 2022). When lower-income households have more children but invest less in each child's education, inequality is transmitted more strongly across generations. In the present model, however, fertility differentials across education groups reverse following the subsidy reform. As a result, intergenerational mobility increases more under endogenous fertility.

While the greater college enrollment in aggregate increases the aggregate productivity and output, the per-capita output increases more under exogenous fertility (+4.0%) than under endogenous fertility (1.3%). Particularly, the equilirium capital stock is more scarce under endogenous fertility since the fertility is high and demographic structure is such that younger households, who have less assets, consist of more part of the economy. Due to the significant increase in per-capita output, the equilibrium tax becomes lower (-0.47 p.p.) than the initial steady state under exogenous fertility, while it becomes slightly higher under endogenous fertility (+0.11 p.p.). Still, the welfare gain—measured as consumption equivalence for new born agents under the veil of ignorance—is greater under endogenous fertility (+6.4%) than exogenous fertility (+3.6%) because of greater mobility in education and income.[30]

Transitional dynamics: Having examined the long-run effects of introducing the income-tested subsidy, I next investigate the transitional dynamics to understand how the economy reaches the new equilibrium. The policy, together with the associated increase in

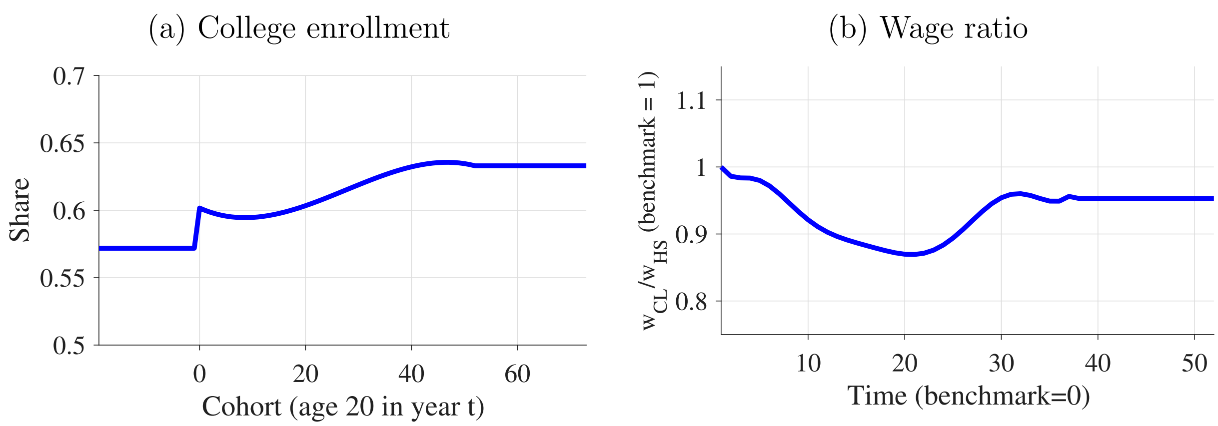
[30]Online Appendix G.1 provides a formal definition of the welfare measure.

the labor income tax rate to its level in the new stationary equilibrium, is implemented unexpectedly and permanently. Since this tax change may not be sufficient to balance the government budget along the transition path, the government adjusts the lump-sum tax accordingly. Households subsequently make optimal decisions along the transition path, taking into account the evolution of prices and tax rates.

The key takeaway is that the macroeconomic effects of the policy take a long time to materialize, partly because fertility responses amplify its impact across generations. For example, Figure 7 (a) shows the transition path of college enrollment rates. Although enrollment rates within cohorts increase immediately following the policy introduction, they do not jump to their new steady-state level and instead converge gradually over time. This is because part of the long-run increase in college enrollment arises from composition effects: students who benefit from the policy become parents in the future, and their children are more likely to attend college due to the intergenerational persistence of assets, abilities, and preferences.

Fertility responses further reinforce these composition effects. In particular, college graduates increase their fertility, and these additional children are themselves more likely to become college graduates. As a result, the effects of the policy are transmitted to subsequent generations, further raising college enrollment over time. The transition is also non-monotonic. As shown in Figure 7 (b), the relative wage of college graduates declines initially because the policy increases the supply of college-educated labor. The resulting decline in the relative market wage of college graduates weakens the incentive to attend college, leading to a temporary slowdown—and even a slight reversal—in the increase in college enrollment rates. Over time, however, the composition effects and fertility channel discussed above become increasingly important, leading to a further rise in college enrollment and eventual convergence to the new long-run equilibrium.

Figure 7: Transitional dynamics of college enrollment rates and wage ratio



Note: In Panel (a), cohort t denotes the cohort that is age 20 in year t and makes its college decision, where $t = 0$ marks the introduction of the policy. Negative (positive) cohort values correspond to cohorts that made their education decisions before (after) the policy was introduced. Panel (b) represents the market wage ratio between college graduates and high school graduates. One model period corresponds to four years.

Expansion: While the previous experiments show that the introduction of the income-tested college subsidy increases aggregate income and welfare in the long run, do these aggregate effects persist under a further expansion of the subsidy? To answer the question, I examine a marginal expansion of the policy by increasing the income threshold $\bar{I}$ in the grant function to the 60th percentile, thereby extending eligibility to students from the third income quintile. Table G.3 reports the quantitative results in the stationary equilibrium, which can be summarized as follows.

First, the expansion leads to a 0.2% decrease in per-capita output, primarily because it reduces incentives to save and thus lowers aggregate capital accumulation in the stationary equilibrium. Second, owing both to lower output and to the broader coverage of the grant program, the equilibrium labor income tax rate increases by 0.92 p.p., substantially more than under the current policy (0.11 p.p.). Third, the aggregate college enrollment rate increases by 5.1 p.p., which is smaller than under the current policy (6.2 p.p.). The main reason is that the higher labor income tax rate weakens the incentive to acquire higher skills. Finally, although aggregate fertility increases further, the effect is fairly small, partly because the higher tax rate suppresses fertility through a negative income effect.

7 Concluding Remarks

This paper studies income-tested education subsidies as a policy that provides insurance against income risk in fertility decisions. The main finding is that income-tested college subsidies increase fertility by reducing education costs contingent on bad income realizations, thereby lowering the expected utility cost of having children under income uncertainty. This insurance channel is quantitatively important and amplifies some of the policy's key long-run effects, including increases in educational attainment and intergenerational mobility.

While this study focuses on college education subsidies, the insurance effect for fertility also applies to income-tested subsidies at earlier education levels and to a broader set of child-related transfers, although its quantitative importance and the resulting macroeconomic implications likely differ across policies. Exploring these broader policies would require extending the current model to incorporate human capital accumulation before college, which this study abstracts from.[31] This paper also abstracts competition for college admissions, which can be important to consider when evaluating broader education policies, such as education taxes and changing enrollment capacities.[32] Incorporating these factors into the current model and exploring broader child-related and education policies are left for future research.

[31] Examples of recent works in this respect, although abstracting from fertility choices, include Lee and Seshadri (2019); Yum (2023); Moschini and Tran-Xuan (2023); Adamopoulou, Hannusch, Kopecky, and Obermeier (2025); Krueger, Ludwig, and Popova (2025); Daruich (2026).

[32] See, for example, Kim, Tertilt, and Yum (2024); Kang (2024); Gu and Zhang (2024).

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Appendix

A Model

A.1 Proof of Proposition 2

Proof. Recall that the first-order condition with respect to n is

$$b'(n(\varepsilon)) = \mathbb{E}\left[a(I;\varepsilon)V'(c(I;\varepsilon))\right],$$

where

$$a(I;\varepsilon) = \underline{a} + \varepsilon(I - \bar{I}),$$

and

$$c(I;\varepsilon) = I - n(\varepsilon)a(I;\varepsilon).$$

Differentiating both sides with respect to ε gives

$$b''(n(\varepsilon))n'(\varepsilon) = \mathbb{E}\left[\frac{\partial a(I;\varepsilon)}{\partial \varepsilon}V'(c(I;\varepsilon)) + a(I;\varepsilon)V''(c(I;\varepsilon))\frac{\partial c(I;\varepsilon)}{\partial \varepsilon}\right].$$

Since

$$\frac{\partial a(I;\varepsilon)}{\partial \varepsilon} = I - \bar{I}$$

and

$$\frac{\partial c(I;\varepsilon)}{\partial \varepsilon} = -n'(\varepsilon)a(I;\varepsilon) - n(\varepsilon)\frac{\partial a(I;\varepsilon)}{\partial \varepsilon},$$

evaluating at $\varepsilon = 0$ yields

$$b''(n(0))n'(0) = \mathbb{E}\left[(I - \bar{I})V'(c_0) + \underline{a}V''(c_0)\left\{-\underline{a}n'(0) - n(0)(I - \bar{I})\right\}\right],$$

where

$$c_0 = I - \underline{a}n(0).$$

Rearranging terms gives

$$\left[b''(n(0)) + \underline{a}^2\mathbb{E}V''(c_0)\right]n'(0) = \mathbb{E}\left[(I - \bar{I})\left\{V'(c_0) - \underline{a}n(0)V''(c_0)\right\}\right].$$

The coefficient on the left-hand side is strictly negative because $b'' < 0$, $\underline{a} > 0$, and $V'' < 0$.

It remains to sign the right-hand side. Define

$$H(I) = V'(c_0) - \underline{a}n(0)V''(c_0).$$

Since $c_0 = I - \underline{a}n(0)$, we have

$$H'(I) = V''(c_0) - \underline{a}n(0)V'''(c_0) < 0,$$

because $V'' < 0$, $V''' > 0$, $\underline{a} > 0$, and $n(0) > 0$. Thus, $H(I)$ is decreasing in I.

Since $I - \bar{I}$ is increasing in I, $H(I)$ is decreasing in I, and $\mathbb{E}(I - \bar{I}) = 0$, it follows that

$$\mathbb{E}\left[(I - \bar{I})H(I)\right] = \mathrm{Cov}(I - \bar{I}, H(I)) < 0.$$

Therefore, the right-hand side is strictly negative, while the coefficient multiplying $n'(0)$ is also strictly negative. Hence,

$$n'(0) > 0.$$

This proves that the introduction of an income-tested subsidy increases fertility. □

A.2 Decomposing the direct effects

While Section 3 suggests that the fertility effects of income-tested subsidies operate through the insurance channel, the subsidy function considered there takes a special form and rules out the price effect. This section therefore derives a decomposition of the fertility effects that incorporates the price effect when the grant function takes a more general form and has a nonzero expected value. To this end, consider a grant function $\bar{g}(I; g, \tau)$, which takes the form

$$\bar{g}(I; g, \tau) = \begin{cases} g + \tau & \text{if } I < \bar{I} \\ \tau & \text{otherwise} \end{cases} \tag{11}$$

The only difference from the one considered in Section 6.2 is the addition of the universal component, τ, which every student can receive regardless of household income. Here, the income threshold $\bar{I}$ and the income-tested component g are set equal to those considered in Section 6.2. The actual policy considered in Section 6.2 corresponds to the special case with $\tau = 0$.

Now, let $n(g, \tilde{\tau}(\cdot))$ denote aggregate fertility under an income-tested component g and the universal grant schedule $\tilde{\tau}(\cdot)$. I define $\tilde{\tau}(\cdot)$ as the household-specific universal grant schedule that assigns to each household the expected value of the income-tested component of the college grant conditional on household income at age J_F:

$$\tilde{\tau}(I_{J_F}) = \mathbb{E}[\bar{g}(I_{J_{IVT}}; g, 0) \mid I_{J_F}],$$

where the expectation is taken over income at age J_{IVT}, when children make college enrollment choices. Under the grant scheme $(g, \tilde{\tau}(\cdot))$, students receive the income-tested component g if parental income at age J_{IVT} falls below the threshold. In addition, they receive the household-specific universal component $\tilde{\tau}(I_{J_F})$ regardless of parental income at age J_{IVT}. Thus, for students whose parents have income I_{J_F} at age J_F, the resulting grant function is $\bar{g}(I; g, \tilde{\tau}(I_{J_F}))$. The percentage change in aggregate fertility in response to the introduction of the actual policy is approximated by $\ln(n(g, 0)) - \ln(n(0, 0))$, which can be rewritten as

$$\underbrace{\ln\left(\frac{n(g,0)}{n(0,0)}\right)}_{=\text{total effect}} = \underbrace{\ln\left(\frac{n(0,\tilde{\tau}(\cdot))}{n(0,0)}\right)}_{=\text{price effect}} + \underbrace{\ln\left(\frac{n(g,0)}{n(0,\tilde{\tau}(\cdot))}\right)}_{=\text{insurance effect}} \tag{12}$$

In the decomposition analysis, I construct a household-specific universal grant scheme without the income-tested component, corresponding to $\bar{g}(I; 0, \tilde{\tau}(\cdot))$, and introduce it in the benchmark model. The resulting aggregate fertility response corresponds to the price effect in (12). It captures the reduction in the expected monetary cost of children's education while abstracting from the insurance value of state-contingent educational support. The insurance effect is then obtained as the difference between the total effect and the price effect. It captures the fertility response arising from the insurance value of the subsidy, holding expected education costs constant.

B Calibration

B.1 Stochastic income process

I estimate the AR(1) process separately for college and non-college graduates, following previous studies that estimate the process using household-level panel data (Daruich and Kozlowski, 2020; Krueger et al., 2025; Daruich, 2026). I first construct a measure of household-level wages, defined as the sum of the husband's and wife's labor income divided by the sum of their working hours. I then compute four-year averages of this household wage measure and estimate a first-stage regression to obtain the residuals. Specifically, separately for college and non-college households, I regress log wages on a quadratic in age, time dummies, and dummies for more detailed education categories within the non-college group (e.g., high school graduates, high school dropouts, etc.). I then obtain the residual for household i of age j and education level e, denoted by $u_{i,j,e}$, which is assumed to consist of two components as follows:

$$u_{i,j,e} = m_{i,j,e} + z_{i,j,e}.$$

The measurement error $m_{i,j,e}$ is normally distributed with mean 0 and variance σ_e^m, and $z_{i,j,e}$ is a persistent shock that follows an AR(1) structure:

$$\begin{aligned} z_{i,j,e} &= \rho_{z,e} z_{i,j-1,e} + \epsilon_{i,j,e} \\ \epsilon_{i,j,e} &\sim N(0, \sigma_{z,e}^2). \end{aligned}$$

The initial draw for z, $z_{i,0,e}$, follows $N(0, \sigma_{z_0,e}^2)$. I estimate this process for each education group using a Minimum Distance Estimator, using the variance of income residuals, their first-order autocovariance, the variance of their first differences, and the initial-period variance of income residuals as moments.

For the sample selection, I first restrict the sample to households whose average weekday hours are between 2 and 24 hours (i.e., on average between 1 and 12 hours per person per weekday), which roughly corresponds to the criteria adopted in the literature (Krueger et al., 2025; Daruich, 2026). I then trim the data to the 1st–99th percentiles of the household wage distribution. In addition, I restrict the sample to households with 24 consecutive observations to ensure sufficiently long four-year intervals for measuring wage variation. Finally, as in Daruich and Kozlowski (2020), I exclude observations reporting large income changes, defined as growth rates above 50% or below −50% when comparing consecutive four-year averages.

B.2 Parameters

Table A.1: Parameters Governing the Age Profile and Stochastic Process of Wages

	High school	College
Age	$+0.041$	$+0.048$
$\text{Age}^2 \times 10,000$	-4.551	-5.364
$\rho_{z,e}$ (persistence)	0.938	0.965
$\sigma_{z,e}$ (volatility)	0.036	0.050

Note: "College" refers to households in which either the husband or the wife is a college graduate. "High school" refers to all other households. The reported age-profile parameters are annual values, which are translated into their four-year counterparts when solving the model. The income risk parameters are estimated using four-year measures of household-level wages, as described in the main text.

Table A.2: Externally Determined Parameters

Parameter	Value	Description	Source
Parameters taken from the literature			
χ	0.70	Elasticity of substitution	Kawaguchi and Mori (2016)
τ_a	0.50	Capital income tax rate	Gunji and Miyazaki (2011)
τ_w	0.33	Labor income tax rate	Gunji and Miyazaki (2011)
Parameters directly measured from data			
α	0.31	Capital share	JIP Database
$\underline{A}$	3.60	Borrowing limit	Family Income and Expenditure Survey
$\underline{A}_s$	0.52	Student borrowing limit	Student Life Survey
p	0.31	Pension benefits	Japan Pension Service
$\bar{t}$	0.21	University-related time	Student Life Survey
τ_c	0.10	Consumption tax rate	Statutory rate in Japan
Parameters estimated from data			
$\{\rho_{z,e}\}_e$	See Table A.1	Income persistence	Japanese Panel Survey of Consumers
$\{\sigma_{z,e}\}_e$	See Table A.1	Income volatility	Japanese Panel Survey of Consumers
$\{\gamma_{j,e}\}_{j,e}$	See Table A.1	Age-earnings profile	Japanese Panel Survey of Consumers

Note: The values corresponding to borrowing limits, pension benefits, and cash transfers are expressed relative to average household income at age 28.

Table A.3: Internally Determined Parameters

Parameter	Value	Description
ι	0.35	Borrowing wedge
ι_s	0.26	Borrowing wedge for students
p_{CL}	0.76	College tuition fees
β	0.82	Time discount factor (four-year)
μ	0.55	Preference weight on consumption
λ_q	0.62	Utility weight on child expenditures
λ_a	0.77	Altruistic discount factor
γ	0.51	Curvature of utility from consumption and leisure
γ_q	0.37	Curvature of utility from child expenditures
γ_n	0.34	Curvature of child discounting
ρ_h	0.73	Intergenerational persistence in ability
σ_h	0.39	Dispersion of ability shocks
ψ_{CL}	342.9	Scale parameter for schooling tastes
ν	2.63	Education sorting by ability
ω	1.95	Intergenerational mobility of education
α_{CL}	0.73	Intercept for education returns
β_{CL}	0.56	Curvature for returns to education
κ	0.12	Time cost of children
Z	3.23	Total factor productivity
δ	0.14	Capital depreciation rate
ω_{CL}	0.52	Weight on college graduates in market production
ψ	0.06	Lump-sum transfers
A_s	0.37	Public spending on higher education

B.3 Validation Exercise: The Benefit Elasticity of Fertility

Many of the existing empirical studies report that the benefit elasticity of fertility—the percentage increase in fertility rate against the one percent increase in the cash transfer—is about $0.1 - 0.2$. For example, Milligan (2005) studies a reform of Quebec's baby bonus and shows that an extra 1,000 Canadian dollars benefit would increase fertility by 16.9%, which implies a benefit elasticity of 0.107. Cohen et al. (2013) uses a variation in the child subsidy payment observed around 2003 in Israel, providing a larger subsidy for third or higher births. They show that the benefit elasticity of fertility was 0.176. Also, González (2013) adopts the regression discontinuity design to study the effects of Spain's reform in 2007, introducing a one-time payment of 2,500 euros (about 3,800 USD) for births, almost 4.5 times the monthly minimum wage for full-time workers. It finds a statistically significant impact on fertility, increasing conceptions by $5 - 6\%$.

To examine how the model performs along this dimension, I conduct the following exercise. Let B_0 denote the per-child cash transfer for households with children under age 20 in the benchmark economy. I solve the household problem while holding prices, the tax rate, and the distribution fixed, and consider several alternative levels of the transfer, $B = B_0(1 + x)$, where $x \in \{1, 2, 3\}$. This range represents a reasonably sized expansion in the context of the policies examined in the empirical literature. For each value of x, I compute the implied fertility response and the corresponding benefit elasticity of fertility.

Table A.4: Targeted Moments: Data vs. Model

Moment	Data	Model
College Enrollment		
Enrollment rate across income quintiles	See Figure 2 (e)	
Enrollment rate by parents' education	See Figure 4	
Fertility		
Fertility across income quintiles	See Figure 2 (c)	
Fertility ratio (college- to high-school-graduates)	0.91	0.94
Distribution of the number of children	See Figure 3	
Parental Transfers		
Average income share of IVTs (students)	0.61	0.73
IVTs across income quintiles	See Figure 2 (f)	
Pre-college spending across income quintiles	See Figure 2 (d)	
Income Inequality		
Skill premium	0.31	0.32
Variance of log wages for college graduates at age 28	0.14	0.15
Intergenerational income elasticity	0.30	0.37
Var(log disposable income)/Var(log gross income)	0.70	0.81
Income levels across income quintiles at age 52	See Figure 2 (a)	
Labor Supply and Saving		
Hours across income quintiles	See Figure 2 (b)	
Average income share of loans (students)	0.21	0.26
Macro & Government		
Interest rate (four-year)	20%	20%
Market wage rates (normalization)	1.0	1.0
Public spending on higher education / GDP	0.003	0.003

The resulting elasticities are 0.066, 0.076, and 0.054, respectively. Taking the average yields an elasticity of 0.065. These estimates are broadly consistent with the empirical evidence and are also conservative, in the sense that the model's response to financial incentives for fertility is no stronger than that typically reported in the literature.

C Empirical Analysis

In Section 6, I show that eliminating income uncertainty increases fertility and that income-tested college subsidies provide partial insurance against income risk in fertility decisions. This section presents suggestive empirical evidence supporting the core mechanism—that income uncertainty reduces fertility—by exploiting variation across contract types, following a strand of the empirical literature that uses employment characteristics as proxies for perceived income uncertainty. I begin by reviewing the empirical literature on the relationship between income uncertainty and fertility. I then provide institutional background and summary statistics by contract type in my sample based on the JPSC. Finally, I estimate regressions examining the association between contract types and birth probabilities and discuss how the results can be interpreted through the lens of income uncertainty.

Literature: Providing direct evidence on the relationship between income risk and fertility is often challenging, primarily because it is difficult to construct a direct measure of perceived income uncertainty—as opposed to realized income shocks—and to obtain plausibly exogenous variation in such a measure. Nevertheless, previous empirical studies examine the link between income uncertainty and fertility in at least two ways.

The first strand of the literature relies on aggregate measures of economic uncertainty and documents a negative association between uncertainty and fertility, typically exploiting cross-country or cross-state variation in measures of uncertainty. For example, Gozgor et al. (2021) use cross-country panel data with the World Uncertainty Index (WUI), compiled by the International Monetary Fund, and find that higher uncertainty is associated with lower fertility. It is important to note, however, that the WUI reflects several dimensions of uncertainty, such as those related to domestic policy and international trade, and does not directly capture income uncertainty. Relatedly, Comolli and Vignoli (2021) focus on Italy's sovereign debt crisis of 2011–2012, construct a measure of uncertainty using Google Trends data for the term "spread," and show—based on a regression discontinuity design—that an increase in perceived uncertainty is associated with lower birth rates. A notable exception that measures income uncertainty more directly is Chabé-Ferret and Gobbi (2018). They exploit variation across U.S. states and over the course of the twentieth century in income uncertainty, measured by the volatility of income growth across states and years. Their results indicate that income uncertainty reduces cohort fertility and that the decline in uncertainty accounts for a significant fraction of the post–World War II baby boom.

The second strand of the literature exploits variation in contract types, which is informative about differences in layoff probabilities, as well as policy reforms that affect workers with different contract types (e.g., Prifti and Vuri, 2013; De Paola et al., 2021). For example, De Paola et al. (2021) exploit a labor market reform in Italy that reduced job security for workers with open-ended contracts and show that the reform reduced fertility among the treated workers. It is important to note, however, that such reforms are likely to affect not only income volatility (a second-moment effect) but also expected income levels (a first-moment effect). Another set of studies along these lines uses variation in contract types—informative about job security—to examine their association with fertility (Modena et al., 2014; Guner et al., 2024). For example, Guner et al. (2024) examine differences between permanent and temporary contracts in Spain and show that temporary workers—who face higher layoff probabilities—are less likely to have a first child.

To provide supporting evidence for my model-based results, I use Japanese data and follow the empirical approach of Guner et al. (2024). The institutional setting is similar to that of Spain studied in Guner et al. (2024). Although variation across contract types in principle captures factors other than exogenous variation in income uncertainty, I carefully specify the empirical framework to better isolate the role of income uncertainty.

Institutional background and descriptive statistics: The Japanese labor market is characterized by a dual structure consisting of two main contract types: regular and non-regular employment. This dual structure originated in Japan's postwar employment system, under which regular workers were granted strong job protection and seniority-based wages. Following the economic slowdown in the 1990s, firms increasingly relied on non-regular workers as a flexible margin of adjustment, a trend facilitated by a series of labor market reforms that expanded the permissible use of fixed-term and dispatched workers. As a result, differences in job security between regular and non-regular workers became institutionalized in the Japanese labor market.

Regular and non-regular contracts differ along several dimensions, including hourly wages, working hours, and layoff probabilities. Regular contracts offer higher wages and stronger job security but require longer working hours with less flexible schedules. In contrast, non-regular contracts offer greater flexibility with fewer working-hour commitments, but are associated with lower pay and weaker employment protection. Women are more likely to work under non-regular contract, while men (particularly married men) are mostly work under regular contracts.

To illustrate these differences, I focus on married women born in 1959–69 in the JPSC. Table A.5 summarizes descriptive statistics on non-regular employment, including the share of non-regular workers by age and education, separation rates, hourly wages, working hours, and birth probabilities.

On average, 24.6% of married women are employed under non-regular contracts, conditional on participation. The share is lower—about 15%—among those under 30 and rises with age. In this cohort, many married women leave the labor force around childbirth and later reenter employment in non-regular jobs, which offer more flexible and less time-demanding work schedules. This lifecycle pattern likely contributes to the increase in non-regular employment with age. Non-regular employment is common among women but not among men: on average, only 2.24% of married men work under non-regular contracts.

As discussed above, non-regular contracts are characterized by (1) higher separation rates, (2) fewer working hours, and (3) lower hourly wages. First, about 3.87% of non-regular workers experience involuntary job loss each year, compared with 0.86% for regular workers. Second, among women aged 30–39, average weekly hours under non-regular contracts are 26.68—about 30% lower than for regular workers (38.13). Third, hourly wages for women aged 30–39 under non-regular contracts are likewise about 30% lower than for their regular-contract counterparts.

Finally, the annual birth probability for non-regular workers is 0.0091, substantially lower than the 0.0792 observed for regular workers. This pattern holds for both college and non-college graduates to a similar extent.

Table A.5: Descriptive statistics on regular and non-regular workers in Japan

	Average	Non-college	College
Share of non-regular workers (conditional on working)			
Women, age < 30	0.1515	0.1540	0.1176
Women, age 30–39	0.2637	0.2795	0.1327
Women, age 40–49	0.2724	0.2817	0.2011
Women, average	0.2464	0.2598	0.1291
Men, average (< 50)	0.0224	0.0306	0.0070
Annual job separation rate			
Regular women	0.0086	0.0092	0.0000
Non-regular women	0.0387	0.0353	0.0909
Weekly working hours (women age 30–39)			
Regular workers	38.13	37.67	40.61
Non-regular workers	26.68	26.93	22.95
Hourly wage (non-regular non-college = 1)			
Regular workers	1.495	1.438	1.805
Non-regular workers	1.025	1.000	1.394
Birth probability			
Regular women	0.0792	0.0786	0.0801
Non-regular women	0.0091	0.0118	0.0026

Note: The sample is restricted to married women born in 1959–1969 and their spauses. Ages are limited to those younger than age 50, which I take as the upper bound of reproductive age. Hourly wages for non-college graduates employed under non-regular contracts are normalized to 1.

Specification: To examine the relationship between non-regular contracts and fertility while controlling for potential confounding factors, I estimate the following linear probability model using the JPSC sample:[33]

$$y_{i,t} = \alpha + \beta T_{i,t-1} + \boldsymbol{x}_{i,t-1}\boldsymbol{\theta} + \psi_i + \phi_t + \varepsilon_{i,t}. \tag{13}$$

Here, $y_{i,t}$ takes one if the wife in a couple i gives birth in year t. $T_{i,t-1}$ is a dummy variable taking one if the wife worked under a non-regular contract in year $t-1$, and

[33]The sign and statistical significance of the coefficient of interest remain unchanged when I estimate the model using a logistic specification, as in Guner et al. (2024).

β is the coefficient of our interest. Using the lagged indicator helps mitigate concerns about reverse causality—namely, that a woman may sort into (or out of) non-regular employment in response to having given birth.[34] The vector $\boldsymbol{x}_{i,t-1}$ includes other couples' characteristics observed in $t-1$, such as household income, household assets, the number of existing children, wife's age, wife's labor force participation, husband's contract type, and *fertility intention*—the number of children she would like to have additionally. ψ_i and ϕ_t denote individual (couple) and year fixed effects. Because the specification includes an indicator equal to one if the woman participated in the labor market in the previous year, the coefficient on the non-regular contract indicator captures the effect of non-regular employment relative to regular employment. In the regression analysis, I focus on married women under age 50 during the period 1993–2019.

Results: Table A.6 reports the regression results. Column (1) presents estimates without fixed effects. The estimate suggests that, holding other variables—such as income, assets, age, number of children, fertility intention, and education—constant, being a non-regular worker is associated with a 2.3 percentage point lower probability of giving birth relative to regular workers. The coefficient is statistically significant at the 1 percent level.

Columns (2) and (3) gradually add couple and year fixed effects. Column (3), which includes two-way fixed effects, implies that within-individual transitions into non-regular employment are associated with a 1.3-percentage-point lower probability of giving birth, conditional on the number of existing children, assets, and fertility intentions. The decline in magnitude suggests that selection based on unobserved individual characteristics partly explains the negative association between non-regular employment and fertility. Nevertheless, the coefficient remains marginally significant, suggesting that selection based on unobserved heterogeneity alone cannot fully account for the observed relationship. This raises the question of which differences between regular and non-regular contract types drive the observed fertility differential.

Table A.6: Regression results for birth probabilities among married women aged 25–49

	(1)	(2)	(3)
Non-regular$_{t-1}$	−0.023***	−0.015**	−0.013*
	(0.006)	(0.007)	(0.007)
Individual FE		✓	✓
Year FE			✓
Observations	5042	5042	5042

* p <0.1, ** p <0.05, *** p <0.01

Note: This table reports the estimated coefficient on an indicator for women who were employed under a non-regular contract in the previous year, predicting the probability of giving birth in the current year. The data are drawn from the JPSC and the sample is restricted to married women under age 50 during 1993–2019. All specifications control for household characteristics, including household income, household assets, the number of existing children, the wife's age, the wife's labor force participation status, the husband's contract type, and fertility intentions, measured as the additional number of children the respondent reports wanting to have.

[34] The results are robust to using a two-year lag of the non-regular employment indicator.

Discussion and interpretation of the results: As discussed earlier, non-regular and regular contracts differ along multiple dimensions, making it unclear which specific component drives the observed relationship. The main differences concern (1) working hours, (2) hourly wages, and (3) job security. Along the working-hours margin, standard fertility models with time costs would predict a positive association between non-regular employment and fertility, as non-regular contracts relax time constraints. This prediction is inconsistent with the negative association between non-regular employment and fertility, suggesting that greater time flexibility is unlikely to be the primary mechanism underlying this relationship.

By contrast, the lower wages and earnings associated with non-regular employment may reduce fertility through a negative income effect, making it difficult to distinguish this mechanism from an income uncertainty channel. To address this concern, I include household income and asset levels as controls. Although these variables do not perfectly capture expected lifetime income—which is arguably the more relevant concept—they partially account for differences in economic resources across contract types. The regression results show that neither income nor asset levels have statistically significant explanatory power for fertility. This finding may not be surprising, given that completed fertility is relatively flat across income quintiles and remains close to two children, as documented in the calibration section.

Taken together, my regression analysis shows that within-individual transitions to less secure contract types are associated with lower birth probabilities, after accounting for time-invariant household characteristics, time-varying fertility intentions, and other controls such as income, assets, age, and the number of existing children. These findings suggest that factors not directly captured in the specification—particularly differences in income uncertainty between regular and non-regular contracts—may constitute an important mechanism underlying the negative association between non-regular employment and fertility.

Online Appendix for

"Education Subsidies as Insurance against Income Risk in Fertility Decisions"

Kanato Nakakuni

D Facts: Education Costs and Fertility Choices in Japan **OA-1**

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D Facts: Education Costs and Fertility Choices in Japan

Japan is a leading country in demographic aging,[35] leading to the shrinking labor force, output, and tax base, while the public expenditures on social security benefits are increasing. As a countermeasure against this demographic issue, the government introduced grants for college students in 2020. Two key underlying premises are: (1) it will increase the "quality" of the labor force in the long run, and (2) it will increase the fertility rate, which increase the "quantity" of the labor force in the long run. This pro-natal motive is explicitly described in an act for introducing the grants.[36]

> The aim is ... to foster an environment where people can bear and raise their children with a sense of ease by alleviating the economic burden associated with higher education, thereby contributing to addressing the rapid decline in the birthrate in our country.
> (Act on Support for Higher Education Studies, enacted on May 17, 2019)

[35] Japan's current fertility rate has been around 1.1, far below the population replacement level. In addition, the old-age dependency ratio is more than 50% and is projected to reach 80% in 2050, which is by far the highest among the OECD countries (See, https://data.oecd.org/).

[36] See, https://www.mext.go.jp/a_menu/koutou/hutankeigen/detail/__icsFiles/afieldfile/2019/05/17/1417025_02_1.pdf (available only in Japanese).

Although that expectation for education policies as a pro-natal policy is unconventional, there are facts suggesting that the financial costs for parents to support their children's college enrollment are a significant barrier to fertility in Japan. I first list the five facts below and then elaborate on each one by one:

1. Japan is one of the least in subsidizing tertiary education.
2. Average parental transfers to college students amount to 65% of the average total education expenditures up to high school graduation.
3. About 80% of parents desire a college education for their children.
4. Couples stop at fewer children than their ideal number primarily due to financial costs.
5. Children's college enrollment is associated with a higher likelihood of parents stopping at fewer children due to the financial reason.

Fact 1. *Japan is one of the least in subsidizing tertiary education.*

Japan is one of the least in subsidizing tertiary education, which roughly corresponds to four-year college education in Japan, given that the enrollment rates for tertiary education other than the four-year college are significantly lower than four-year college enrollment rate.[37]

To see this, I use the cross-country data provided by the OECD[38] and define the *subsidization rate* for a specific education category e as follows:

$$s_e = \frac{y_e^{pub}}{y_e^{pri} + y_e^{pub}},$$

where y_e^{pri} and y_e^{pub} represent the private and public spending on education e, both represented as a share of the GDP. Public spending includes expenditures on educational institutions and educational-related subsidies for households or students. Private spending refers to expenditures financed by households and other private entities. Private spending includes expenditures on school but excludes those outside educational institutions (e.g., textbooks purchased by households, private tutoring, and student living costs).[39] In other words, the subsidization rate indicates what fraction of (potential) school-related costs are funded by the government.

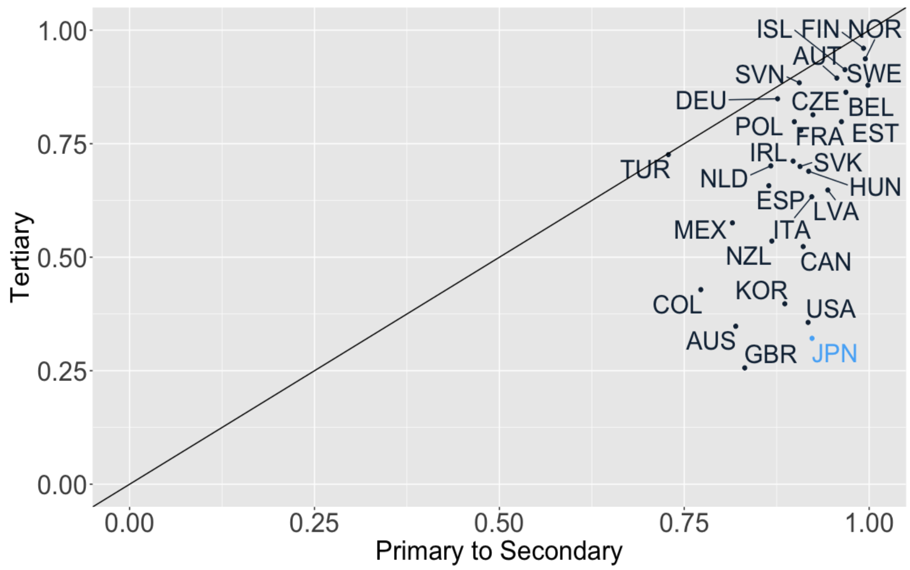
As Table D.1 shows, Japan subsidizes more than 90% of the costs for primary-to-secondary education, which is higher than the OECD average. On the contrary, the subsidization rate for tertiary education was only 32%, which is the second lowest among OECD countries and less than half of the OECD average of 70%. This fact shows plenty of room for increasing subsidies for college students, which also has driven recent policy discussions on introducing and expanding education subsidies for college students.

[37]According to the Basic School Survey by the MEXT, the enrollment rate for some college was 4% in 2022 and is declining over the past thirty years.

[38]I use 2018's data given that it is the latest year in which data on a significant number of countries is available.

[39]For more details, see the OECD data (https://data.oecd.org/).

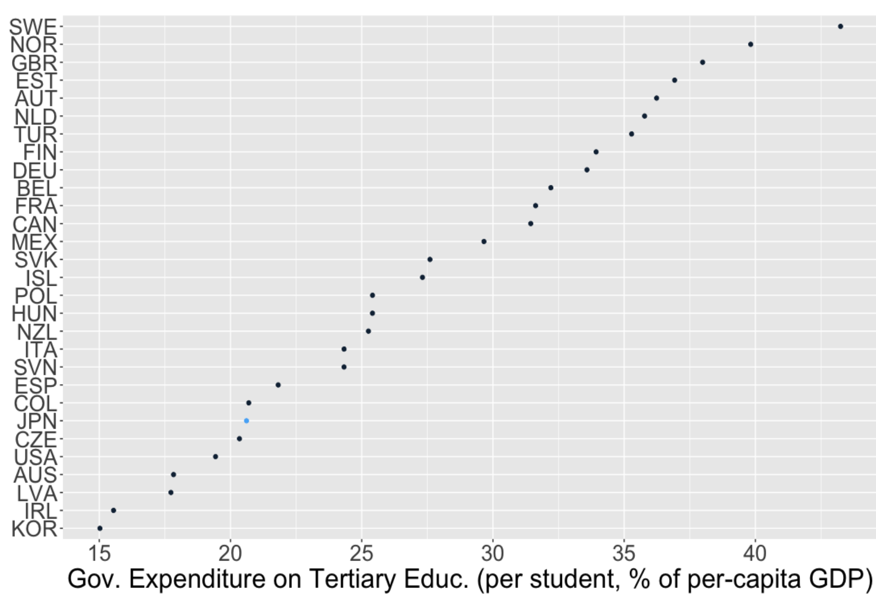
Figure D.1: Subsidization rate for each education category



Note: The data come from the OECD (2018). The subsidization rate is defined as the share of public expenditure in total education expenditure, where total expenditure is the sum of public and private expenditure. Public expenditure includes spending on educational institutions as well as education-related subsidies to households and students. Private expenditure refers to spending financed by households and other private entities.

Figure D.2 plots the ratio of per-student government expenditures on tertiary education to per-capita GDP. On average, the government expenditures for each student enrolling in tertiary education amount to 33% of per-capita GDP, but the Japanese government only does to 21%.

Figure D.2: Per-student government spending on tertiary education to per-capita GDP



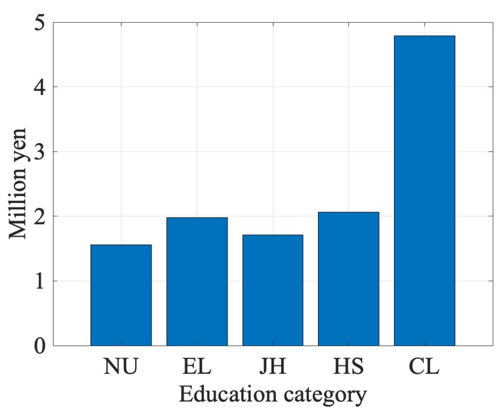
Note: The data come from the OECD (2016–2017).

Fact 2. *Average parental transfers to college students amount to 65% of the average* *total education expenditures up to high school graduation.*

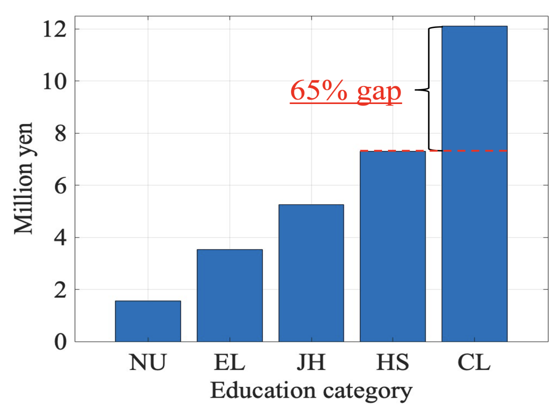
I use two data sets: (1) the Survey of Children's Learning Expenses (SCLE, 2021) and (2) the Student Life Survey (SLS, 2018), both cross-sectional household surveys and conducted by the Ministry of Education, Culture, Sports, Science and Technology (MEXT).[40] The SCLE covers more than 53,000 students from preschool to high school and reports the per-student average education expenditure for each expenditure category and education stage (i.e., preschool, elementary, junior high, and high school). The expenditure category includes not only school-related ones (e.g., tuition fees and textbooks) but also extracurricular activities (e.g., cram school, music, arts, and sports). In addition to that, I use the SLS for parents' expenditures when their children enroll in college.[41]

Figure D.3: Education expenditures

(a) By education stage



(b) Cumulative



Note: The data come from the Survey of Children's Learning Expenses (SCLE, 2021) and the Student Life Survey (SLS, 2018). "NU," "EL," "JH," "HS," and "CL" each stand for nursery school or preschool, elementary school, junior high school, high school, and college, only consisting of a four-year college. All expenditures are conditional on enrollment in the education stage.

Figure D.3 (a) shows the average per-child expenditure for each education category until completion. "NU," "EL," "JH," "HS," and "CL" each stand for nursery school or preschool, elementary school, junior high school, high school, and college, only consisting of a four-year college. All expenditures are conditional on enrollment in the education stage. Figure D.3 (b) represents the cumulative education expenditures, showing that the expenditure jumps up when children attend college. Given that the high school graduation rate is approximately 100% in Japan, the figure tells us that raising one child on average takes the education costs of 7.31 million yen, which amounts to 4.5% of the individuals' average lifetime labor earnings.[42] If their children attend college, they have to

[40] See, `https://www.mext.go.jp/b_menu/toukei/chousa03/gakushuuhi/1268091.htm` for the SCLE and `https://www.jasso.go.jp/statistics/gakusei_chosa/__icsFiles/afieldfile/2021/03/09/data18_all.pdf` (in Japanese) for the SLS.

[41] In 2018, the SLS correct answers from 43,394 students attending tertiary education, including college, some college, and graduate school.

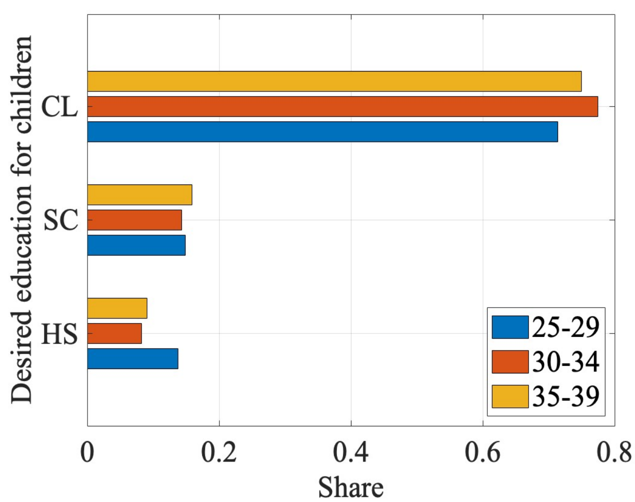
[42] I construct a proxy of the average individual's lifetime earnings based on the 2022 Basic Survey on Wage Structure (BSWS) by the Ministry of Health, Labour, and Welfare (MHLW). First, I compute the average monthly earnings of ordinary workers for each age unconditional on any other characteristics

spend another 4.8 million yen, meaning there is about a 65% increase in education costs if parents send their children to college. This expensiveness of college education can, at least partly, be attributed to the fact that Japan is one of the least in subsidizing tertiary education while subsidizing a sizable portion of schooling costs up to secondary education. These observations establish Fact 2: A significant financial cost gap exists between those who have children enrolled in college and those who do not.

Fact 3. *About 80% of parents desire a college education for their children.*

We return to the NFS (2015), which asked respondents about the desired education level for their children. Figure D.4 summarizes the results by wife's age. Here, "SC" stands for some college. The figure shows that approximately 75% of the parents desire a college education for their children, which is significantly higher than the college enrollment rate, which was 56.6% in 2022. This observation suggests that, although college enrollment in Japan has been far below 100%, education costs for a college education can be relevant not only for a specific part of the population because many parents would like to send their children to college.

Figure D.4: Wives' intention for children's education attainment.



Note: The data come from the National Fertility Survey (NFS, 2015). "CL," "SC," and "HS" stand for college, some college, and high school. Bars in different colors represent different age groups of respondents.

Fact 4. *Couples stop at fewer children than their ideal number primarily due to financial costs.*

This fact is drawn from the National Fertility Survey (NFS), which is provided by the National Institute of Population and Social Security Research (IPSS).[43] The NFS is a cross-sectional household survey that asks respondents about their preferences or

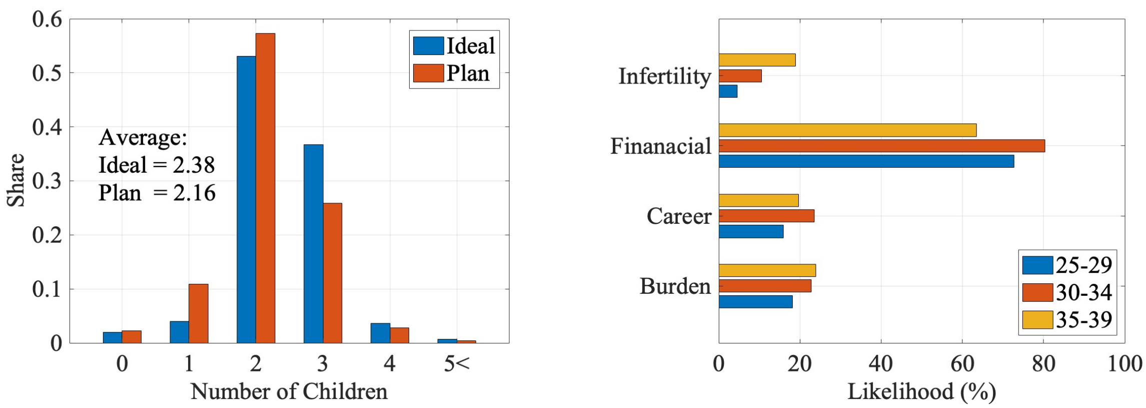
such as sex and education. Then, I sum up the average earnings for each age, which amounts to about 160 million yen, and regard it as a proxy of the individuals' average lifetime labor earnings.

[43]See, https://www.ipss.go.jp/site-ad/index_english/survey-e.asp.

intentions regarding fertility, marriage, child-raising, and education, as well as their basic information, including their education, age, and income. It is conducted nearly every five years, and the latest survey available is in 2015, which collected 5,334 couples in which the wife is aged 18 to 49. Hereafter, I present the results focusing on married couples in which the wife is aged 25 to 39 years, leaving 2,420 couples.[44] According to the NFS, a non-negligible gap exists between the ideal and planned numbers of children. In the 2015 survey, the ideal number of children for wives aged 25 to 39 was on average 2.38, whereas the planned number was 2.16. Figure D.5 (a) represents the distribution of the ideal and planned numbers of children, where blue (red) bars indicate the share of wives who desire (plan) to have each number of children from zero to more than five. This figure suggests that the gap originates from the downward revision of the ideal at the intensive margin. There is no significant share gap between those whose ideal number is zero and those whose planned number is zero, and a substantial fraction of wives who desire three children end up with one or two children.

Figure D.5: Ideal and planned numbers of children

(a) Distribution (b) Reasons for the gap



Note: The data come from the National Fertility Survey (NFS, 2015). In panel (b), bars in different colors represent different age groups of respondents. The reasons reported in panel (b) are defined as follows. *Financial*: "raising children and education are too expensive." *Career*: "it would interfere with my job." *Burden*: "I would not be able to handle the psychological and physical burden" (associated with achieving the ideal number of children). *Infertility*: "not being able to conceive despite wanting to do so."

Why do people abandon their ideal number of children? The NFS asks them to pick their reasons among several options, and the result indicates that they are most likely to choose the financial reason: "raising children and education are too expensive." Figure D.5 (b) represents the likelihood[45] that wives of each age group abandon their ideal number of children for a particular reason, and "Financial" represents the financial reason. Aside

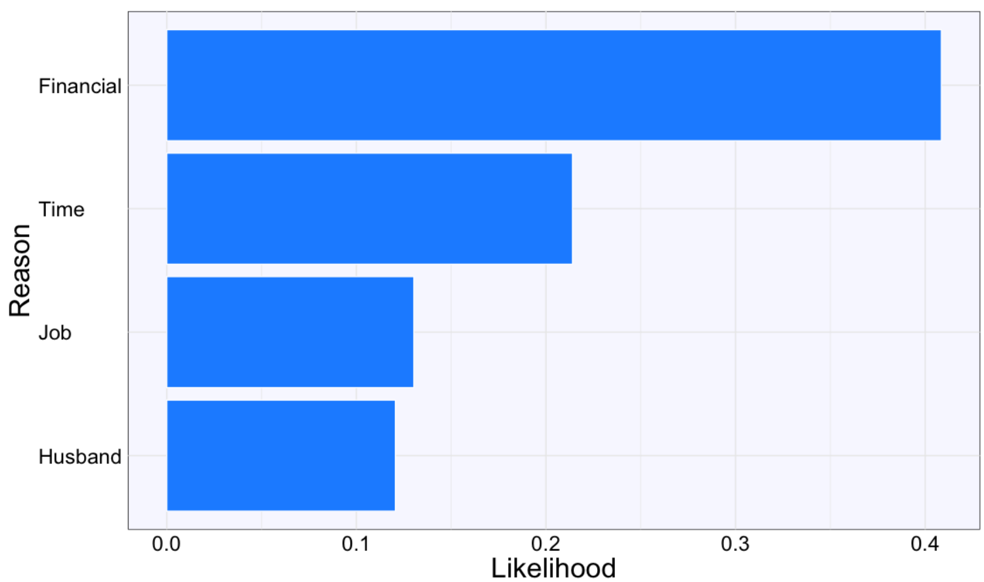
[44]Its 53.1% of the sample consists of couples in which the wife is over age 40, so the sample size shrinks if we target the younger couples. I focus on wives under age 39 because here I am interested in the fertility intention of those in the stage of fertility decision. Note that the cohort fertility rate is stable after age 40 for any cohort in Japan. See, for example, p7 of https://www.mhlw.go.jp/toukei/saikin/hw/jinkou/tokusyu/syussyo07/dl/gaikyou.pdf (in Japanese). Excluding those aged 18 to 24 does not affect the result significantly because they consist only of 1.5% of the observations for wives aged 18 to 49.

[45]I use the term "likelihood" instead of "share" because it allows respondents to choose multiple options.

from this, “Career” represents “it would interfere with my job,” and “Burden” represents “I would not handle the psychological and physical burden” (arising from achieving the ideal number). On average, more than 75% of them chose the financial reason, which dominates other reasons, such as “Career” and “Burden,” chosen by only 20% of them.

We observe the same even using the JPSC, which asks respondents about their preferences or intentions regarding fertility, such as “Do you want more children in the future?” If they answer “No,” it then asks the reasons from fourteen options, which they can choose multiple. The main options are: (1) having and educating children takes too much financial cost; (2) I would rather spend more time with my husband or for myself; (3) I want to continue my job; and (4) I cannot expect my husband’s effort in child-rearing. Figure D.6 reports the likelihood for each of the four reasons, where I refer to (1), (2), (3), and (4) as *Finanial*, *Time*, *Job*, and *Husband*, respectively. This figure indicates that more than 40% of parents cease reproduction due to the financial reason, which is the most important among other reasons.

Figure D.6: Likelihood for reasons why parents cease reproduction



Note: Each of the four reasons, *Finanial*, *Time*, *Job*, and *Husband*, represent: (1) having and educating children takes too much financial cost; (2) I would rather spend more time with my husband or for myself; (3) I want to continue my job; and (4) I cannot expect my husband’s effort in child-rearing.

To sum up, there is a non-negligible gap between the ideal and planned numbers of children for couples, and a substantial fraction of them answer the reason as “raising children and education are too expensive.” These results establish the first fact: couples are most likely to abandon having an ideal number of children because of financial costs.

Fact 5. *Children’s college enrollment is associated with a higher likelihood of parents stopping at fewer children due to the financial reason.*

This fact links the above four facts and suggests that college education costs are an important determinant of fertility choices in Japan. To see this, I conduct the following regression to examine the relationship between fertility *intentions* and children's college enrollment:

$$\Pr(y_i = 1) = L(\alpha + \beta \mathit{ChildCL}_i + \boldsymbol{x}_i\boldsymbol{\theta} + \varepsilon_i),$$

Here, y_i is a dummy variable taking one if a couple i decides to stop having an additional child due to the financial reason. The results are summarized in Table D.1.

Table D.1: Children's college enrollment and fertility intentions of women aged 24-39

	(1)	(2)
$\mathit{ChildCL}_i$	0.515^{***}	0.870^{***}
	(0.186)	(0.222)
Personal characteristics	No	Yes
Observations	692	

Note: Values in parentheses represent standard errors. *** denotes statistical significance at the 1 percent level.

The first column presents the estimates where I control only for the children's college enrollment indicator. In the second column, I add couples' characteristics, such as educational backgrounds. The first and second columns report that the log odds ratio for couples choosing the financial reason to stop having children increases by 0.515 and 0.870 points for couples whose children will enroll in college. Both are statistically significant, and the latter implies a 2.4 times higher odds ratio for stopping having an additional child due to financial reasons.

E Quantitative Model

E.1 Value Functions

Early Working Stage: The remaining component of the value functions during the education stage, V^w, represents value functions in the early working stage (before fertility stage) for households aged $j \in \{J_E, \ldots, J_F - 1\}$. The state variables for this stage consists of asset a, age j, idiosyncratic component of labor productivity z, education level e, and innate ability h. The uncertainty in this stage is only about the next period's productivity, which is denoted by z' following a Markov process $\pi_z(z', z)$. Households choose consumption, leisure, and savings given the state variables. The value function is formulated as follows:

$$V^w(a,j,z;e,h) = \max_{c,l,a'}\{u(c,l) + \beta \mathbb{E}_{z'}\mathcal{V}'\},$$

$$\mathcal{V}' = \begin{cases} V^f(a',z',e,h) & \text{if } \ j = J_F - 1 \\ V^w(a',j+1,z';e,h) & \text{otherwise} \end{cases}$$

s.t.

$$(1+\tau_c)c + a' = (1-\tau_w)w_e\eta_{j,e,h,z}(1-l) + \psi + \tilde{a}(a),$$

$$z' \sim \pi(z',z), \ \ a' \geq -\underline{A},$$

$$\tilde{a}(a) = \begin{cases} (1+(1-\tau_a)r)a & \text{if } a \geq 0 \\ (1+r^-)a & \text{otherwise} \end{cases}$$

Value functions V^f and V^r represent those for the fertility stage and retirement stage, which are formulated in Section 4.3.

Late Working Stage: At this stage for households aged $j \in \{J_{IVT}, ..., J_R - 1\}$, the value function is formulated as:

$$V^w(a,j,z;e,h) = \max_{c,l,a'}\{u(c,l) + \beta \mathcal{V}'\},$$

$$\mathcal{V}' = \begin{cases} V^r(a',j+1) & \text{if } \ j = J_R - 1 \\ \mathbb{E}_{z'}[V^w(a',j+1,z';e,h)] & \text{otherwise} \end{cases}$$

s.t.

$$(1+\tau_c)c + a' = (1-\tau_w)w_e\eta_{j,e,h,z}(1-l) + \psi + \tilde{a}(a),$$

$$z' \sim \pi(z',z),$$

$$a' \geq \begin{cases} 0 & \text{if } \ j = J_R - 1, \\ -\underline{A} & \text{otherwise.} \end{cases}$$

$$\tilde{a}(a) = \begin{cases} (1+(1-\tau_a)r)a & \text{if } a \geq 0 \\ (1+r^-)a & \text{otherwise} \end{cases}$$

Here, V^r denotes the value function for the retirement stage, as described below.

Retirement Stage: At the beginning of age J_R, households retire from the labor market. After that, they spend all their time on leisure and make consumption-saving decisions. Two additional points differ from previous choice problems: they receive the pension benefit p from the government and face uncertainty about the next period's survival. The value function for the retirement stage is formulated as follows:

$$V^r(a,j) = \max_{c,a'} u(c,1) + \beta \xi_{j,j+1} V^r(a',j+1)$$

s.t.

$$(1+\tau_c)c + a' = p + (1+(1-\tau_a)r)a + \psi,$$

$$a' \geq 0.$$

E.2 Equilibrium Definition

Let $\boldsymbol{x}_j^e$ be an age-specific state vector for agents with education level $e \in \{HS, CL\}$ and $\mu_j^e(\boldsymbol{x}_j^e)$ be the measure of agents with state vector $\boldsymbol{x}_j^e$. Let $I_l(h, I)$ and $I_g(h, I)$ be indicator functions for loans and grants, respectively, returning 1 if students with (h, I) are eligible and 0 otherwise.

Given exogenous parameters and policy rules $\{\tau_a, \tau_c, \iota, \iota_s, B, S, \psi, p, I_l(h, I), I_g(h, I)\}$, a *stationary recursive competitive equilibrium* consists of

- value functions $\{V_{g0}, V_{g1}, V^w, V^f, V^{wf}, V^{IVT}, V^r\}$,
- policy functions for consumption, savings, leisure $\{c_j^e(\boldsymbol{x}_j^e), a_j^e(\boldsymbol{x}_j^e), l_j^e(\boldsymbol{x}_j^e)\}_{j=J_E}^{J}$, working hours $\{h_j^e(\boldsymbol{x}_j^e)\}_{j=J_E}^{J_R}$, fertility $\{n_{J_F}^e(\boldsymbol{x}_{J_F}^e)\}$, IVT $\{a_{J_{IVT}^e}(\boldsymbol{x}_{J_{IVT}}^e)\}$, and college enrollment $\{e_{J_E}(\boldsymbol{x}_{J_E})\}$,
- prices (r, w_{HS}, w_{CL}),
- labor income tax rate τ_w,
- aggregate quantities (K, L_{HS}, L_{CL}),
- measures for households $\{\mu_j^e(\boldsymbol{x}_j^e)\}_{j=J_E}^{J}$,

such that:

1. The decision rules of students, workers, and retired households solve their problems, and $\{V_{g0}, V_{g1}, V^w, V^f, V^{wf}, V^{IVT}, V^r\}$ are the associated value functions.
2. The representative firm maximizes its profit and optimally chooses capital and labor inputs:

$$r + \delta = \alpha \cdot Z \cdot \left(\frac{K}{L}\right)^{\alpha-1}, \tag{14}$$

$$w_{HS} = \tilde{Z} \cdot \omega_{HS} \cdot L_{HS}^{\chi-1}, \tag{15}$$

$$w_{CL} = \tilde{Z} \cdot \omega_{CL} \cdot L_{CL}^{\chi-1}, \tag{16}$$

   where

$$L = \left[\omega_{HS} \cdot (L_{HS})^{\chi} + \omega_{CL} \cdot (L_{CL})^{\chi}\right]^{1/\chi},$$
$$\tilde{Z} = (1-\alpha) \cdot Z \cdot \left(\frac{K}{L}\right)^{\alpha} \cdot \left[\omega_{HS} \cdot (L_{HS})^{\chi} + \omega_{CL} \cdot (L_{CL})^{\chi}\right]^{1/\chi - 1}.$$

3. The labor market for each skill $e \in \{HS, CL\}$ clears:

$$L_e = \sum_{j=J_E}^{J_R} \int_{\boldsymbol{x}_j^e} \eta_j^e(\boldsymbol{x}_j^e) \cdot h_j^e(\boldsymbol{x}_j^e)\, d\mu_j^e(\boldsymbol{x}_j^e), \tag{17}$$

   where $\eta_j^e(\boldsymbol{x}_j^e)$ represents the labor efficiency of agents with a state vector $\boldsymbol{x}_j^e$.

4. The capital market clears:

$$K = \sum_{j=J_E}^{J} \sum_{e} \int_{\boldsymbol{x}_j^e} a_j^e(\boldsymbol{x}_j^e)\ d\mu_j^e(\boldsymbol{x}_j^e). \tag{18}$$

5. The government budget is balanced:

$$\begin{aligned}\tau_c \cdot C + \tau_w \cdot (w_{HS} L_{HS} + w_{CL} L_{CL}) + \tau_a \cdot r \int_{\boldsymbol{x}} \mathbb{I}_{a(\boldsymbol{x})>0} a(\boldsymbol{x}) dF(\boldsymbol{x}) + Q \\ = p \cdot \mu_{old} + \psi + B \cdot \mu_{j\leq 20} + S + (\iota - \iota_s) \cdot K_s + G + \mu_s \cdot A_s,\end{aligned}$$

where

$$\begin{aligned} C &= \sum_{j=J_E}^{J} \sum_{e} \int_{\boldsymbol{x}_j^e} c_j(\boldsymbol{x}_j^e)\ d\mu_j^e(\boldsymbol{x}_j^e), \\ Q &= \sum_{j=J_R+1}^{J} \frac{1-\zeta_{j-1,j}}{\zeta_{j-1,j}} \sum_{e} \int_{\boldsymbol{x}_j^e} a_j^e(\boldsymbol{x}_j^e)\ d\mu_j(\boldsymbol{x}_j^e), \\ p \cdot \mu_{old} &= \sum_{j=J_R}^{J} \sum_{e} \int_{\boldsymbol{x}_j^e} p\ d\mu_j^e(\boldsymbol{x}_j^e), \\ K_s &= \int_{\boldsymbol{x}^s} \max\{0, -a^s(\boldsymbol{x}^e)\} \cdot I_l(h, I)\ d\boldsymbol{x}^s \\ G &= \int_{\boldsymbol{x}^s} g(h, I) \cdot I_g(h, I)\ d\boldsymbol{x}^s \\ B \cdot \mu_{j<20} &= \sum_{j=J_F}^{J_{IVT}-1} \sum_{e} \int_{\boldsymbol{x}_j^e} B \cdot n\ d\mu_j^e(\boldsymbol{x}_j^e), \end{aligned}$$

where $\boldsymbol{x}^s$, $\mu^s(\boldsymbol{x}^s)$, and $\{a^s(\boldsymbol{x}^s)\}$ represent a state vector for college students, measure of college students, and students' policy function for saving, respectively.

6. Distributions (measures) and households' behavior are consistent.

E.3 Discussion of Model Assumptions

E.3.1 Pre-College Human Capital Formation

In this model, parental expenditures made before their children graduate from high school (q) do not contribute to human capital formation and therefore do not affect the children's initial endowment. One might think that abstracting from endogenous pre-college human capital accumulation could bias the fertility effects of income-tested education subsidies. If college becomes more affordable, parents may increase investments in their children's human capital prior to college enrollment, as higher human capital at college entry raises the returns to higher education. Through the quantity–quality trade-off, this mechanism could attenuate the positive fertility response to the subsidy. Although this may be true for some households, endogenous pre-college human capital accumulation could also strengthen the fertility response for others, and therefore the overall implications for aggregate fertility are not clear-cut.

For households whose children already have a high probability of attending college, the subsidy frees up resources that would otherwise be used to finance college expenditures, which may generate a positive income effect that increases fertility. Indeed, these heterogeneous responses arise in the current analysis: the price effect of the subsidy reduces fertility among less educated households through the quantity–quality trade-off, while it increases fertility among college-educated households through a positive income effect (Section 6.2.1). Moreover, the insurance effect of the subsidy may become even stronger in an environment with endogenous human capital accumulation. In such an environment, bad income shocks during child-rearing years are more costly because reduced parental investments generate persistent consequences through dynamic complementarity in skill formation. At the same time, parents must save to finance future college expenditures, creating a trade-off between investing in children's human capital before college and reserving resources for future college costs. Income-tested education subsidies relax this trade-off by reducing college education costs contingent on bad income shocks, thereby increasing the value of insurance they provide. As a result, the insurance effect may become quantitatively more important when pre-college human capital accumulation is endogenous. Taken together, these considerations suggest that the direction of the bias from abstracting from endogenous pre-college human capital accumulation is not clear-cut.

That said, incorporating endogenous pre-college human capital formation remains an important avenue for future research, as it would allow for the analysis of a broader set of education policies and their implications for fertility decisions and macroeconomic outcomes. This paper takes a first step by studying the insurance effects of college subsidies on fertility and their macroeconomic implications, leaving the extension to endogenous pre-college human capital formation for future work.

E.3.2 Fertility Timing

In this model, the timing of fertility choice is fixed at age J_F. While this assumption may arguably introduce an upward bias in the insurance role of the income-tested subsidy—since households may postpone childbearing under income uncertainty, and excluding this additional adjustment margin could overstate the effect—the direction of the bias is not clear-cut.

Indeed, fixing fertility timing could also lead to a downward bias, as income-tested subsidies may allow households to have children with less reliance on self-insurance. Without such subsidies, households might postpone childbirth until they are confident in their ability to self-insure—typically through accumulated savings. In contrast, when subsidies are available, the need for precautionary savings is reduced, which may encourage earlier childbearing. In this case, the insurance effect of the subsidy becomes more pronounced, as households bear a lower level of self-insurance than they would if they had postponed childbirth.

At the same time, the assumption may still lead to an upward bias for the same reason noted earlier: excluding the option to postpone childbirth may remove an important channel for mitigating the risk associated with having a child. However, reproductive ability declines exponentially with age, and delaying childbirth increases the risk of infertility, especially after age 35 (e.g., Sommer, 2016), which constitutes a cost of postponement. In my JPSC data sample, the average age at first birth is 30.01, suggesting that the marginal cost of postponement in terms of infertility risk is non-negligible. As a result, households have an incentive to avoid delaying childbirth, making income-tested subsidies a valuable form of insurance in such decisions. This cost of postponement is more pronounced for more educated parents, where college graduates giving birth later on average (31.34) than high school graduates (29.75). Since the insurance effect is more significant for college-educated parents, who are more likely to send their children to college, the potential upward bias introduced by fixing the timing of fertility is likely to be less severe than if the insurance effect were more concentrated among high school graduates.

F Computational Algorithm

F.1 Stationary Equilibrium

For any variable or distribution x, let $\tilde{x}$ and $\hat{x}$ represent its guessed and model-implied values. Also, let μ_j represent the distribution over state variables for age j. Dividing the computation process into three blocks makes it easier to understand. The first block is the outer loop, searching for equilibrium prices. The other two blocks are inner loops; one is the *optimization block*, solving household problems given prices, and another is the *distribution block*, searching for the stationary distributions given prices and policy functions obtained in the optimization block. The computational algorithm proceeds as follows:

1. Guess prices $\tilde{\boldsymbol{p}} = (\tilde{r}, \tilde{w}_{HS}, \tilde{w}_{CL})$.

2. *Optimization block*:

   - Guess the value function for agents at the beginning of age $j = 18$ ($\tilde{V}_{g0}$).
   - Given $\tilde{V}_{g0}$, solve backward from the period of IVT choice to that of the education choice, which gives the model-implied value function for V_{g0}, $\hat{V}_{g0}$.
   - Check if
   $$d(\tilde{V}_{g0}, \hat{V}_{g0}) < \varepsilon, \tag{19}$$
   where $d(\cdot)$ and $\varepsilon > 0$ represent an arbitrary metric function and error tolerance. If (19) is not satisfied, update $\tilde{V}_{g0}$ and follow the same procedure until convergence. The correct V_{g0} pins down all value functions and policy functions given a set of prices $\tilde{\boldsymbol{p}}$.

3. *Distribution block*:

   - Guess the distribution for age J_{IVT}. This $\tilde{\mu}_{J_{IVT}}$ and policy functions for IVT derive the implied distribution for agents aged 18, $\hat{\mu}_{18}$. Given $\hat{\mu}_{18}$ and policy functions, compute the implied distributions for age $j = 19, ..., J_{IVT}$, and obtain $\hat{\mu}_{J_{IVT}}$.

- Check if

$$d(\tilde{\mu}_{J_{IVT}}, \hat{\mu}_{J_{IVT}}) < \varepsilon. \tag{20}$$

If (20) is not satisfied, update $\tilde{\mu}_{J_{IVT}}$ and follow the same procedure until convergence.

- After obtaing the correct distributions for age $j = 18, ..., J_{IVT}$, compute the distribution for age $j_{IVT} + 1$ onward, using those distributions and policy functions.

4. After computing value functions, policy functions, and distributions, compute the implied quantities, $\hat{L}$ and $\hat{K}$ based on (17) and (18), which gives the implied prices $\hat{\boldsymbol{p}}$ based on $\hat{L}$, $\hat{K}$, (14), (15), and (16).

5. Check if

$$d(\tilde{\boldsymbol{p}}, \hat{\boldsymbol{p}}) < \varepsilon. \tag{21}$$

If (21) is not satisfied, update $\tilde{\boldsymbol{p}}$, return to the optimization block, and follow the same procedure until convergence.

F.2 Transition Dynamics

For any variable or distribution x, let $\tilde{x}$ and $\hat{x}$ represent its guessed and model-implied value (or distribution). Also, let $\boldsymbol{q}_t = (K_t, L_t)$ be a vector of aggregate quantities in a year t, and $\boldsymbol{q} = (\boldsymbol{q}_t)_{t=1,...,T}$ for some T. The computational algorithm for transition dynamics proceeds as follows:

1. *Preliminaries*: Set an arbitrarily large number for transition periods, T. Given aggregate quantities in the initial and final steady states, $\boldsymbol{q}_1$ and $\boldsymbol{q}_T$, guess a sequence of aggregate quantities, $\{\tilde{\boldsymbol{q}}_t\}_{t=1,...,T}$, where $\tilde{\boldsymbol{q}}_1 = \boldsymbol{q}_1$ and $\tilde{\boldsymbol{q}}_T = \boldsymbol{q}_T$, which pins down guesses for prices $\{\tilde{r}_t, \tilde{w}_t\}_{t=1,...,T}$. Also, guess a sequence of the labor income tax rate $\{\tilde{\tau}_{l,t}\}_{t=1,...,T}$, where $\tilde{\tau}_{l,1}$ and $\tilde{\tau}_{l,T}$ are the tax rates at the initial and final steady states.

2. *Optimization*: Given $\{\tilde{r}_t, \tilde{w}_t\}_{t=1,...,T}$ and $\{\tilde{\tau}_{l,t}\}_{t=1,...,T}$, solve the optimization problem for each cohort born before period T.

3. *Distribution*: Given decision rules for each cohort obtained in the above optimization block, construct the implied distributions:

   - $\tilde{\mu}_a(j;t)$: (aggregate) savings distribution across age in year t.
   - $\tilde{\mu}_n(j;t)$: (aggregate) labor supply distribution across working-age in year t.
   - $\tilde{\mu}_{j|t}$: age distribution in year t.

4. Given the decision rules and the implied distributions, compute the model-implied aggregate quantities, $\hat{\boldsymbol{q}}$. Also, compute the implied government revenue and expenditures, denoted by $\hat{\boldsymbol{G}}_r = (\hat{\boldsymbol{G}}_{r,t})_{t=1,...,T}$ and $\hat{\boldsymbol{G}}_e = (\hat{\boldsymbol{G}}_{e,t})_{t=1,...,T}$.

5. Check if

$$d(\tilde{\boldsymbol{q}}, \hat{\boldsymbol{q}}) < \varepsilon$$

and

$$d(\hat{\boldsymbol{G}}_r, \hat{\boldsymbol{G}}_e) < \varepsilon.$$

If the first (second) condition is not satisfied, update $\tilde{\boldsymbol{q}}$ ($\{\tilde{\tau}_{l,t}\}_{t=1,...,T}$) and follow the same procedure until convergence.

G Counterfactual Experiments

G.1 On the Welfare Analysis

The welfare analysis using models with endogenous fertility is not straightforward theoretically and philosophically. One of the well-known difficulties is that the standard concept of Pareto efficiency is not applicable to models with endogenous fertility (Golosov, Jones, and Tertilt, 2007).[46] While recognizing these difficulties, this study discusses the welfare effects of the policy by computing the consumption equivalence under the veil of ignorance under the benchmark economy relative to the new steady state,[47] which is a useful standpoint for considering the potential channel through which education subsidies affect economic welfare.

To formalize the measure, let $P \in \{0, 1, 2, ...\}$ denote education subsidy schemes or policies with $P = 0$ representing the benchmark economy without grants. Next, let $V^P(\lambda)$ be the expected lifetime utility in stationary equilibrium for newborn agents with a consumption scaling parameter λ and policy P:

$$V^P(\lambda) = \int_{\boldsymbol{x}_1} V^P_{j=1}(\boldsymbol{x}_1; \lambda) d\mu(\boldsymbol{x}_1). \tag{22}$$

$\mu(\boldsymbol{x}_j)$ is the measure over the age-specific state space where $j \in \{1, ..., J\}$ and $V^P_{j=1}(\boldsymbol{x}_1; \lambda)$ represents the expected utility for an agent aged $j = 1$ with a state vector $\boldsymbol{x}_1$:

[46] This is because models with endogenous fertility require a welfare comparison between two different sets of individuals. Golosov, Jones, and Tertilt (2007) then propose two efficiency concepts that can apply to models with endogenous fertility. For example, the $\mathcal{A}-$efficiency proposed by them focuses on individuals being alive in both economies. This concept is particularly beneficial if we examine the optimal policy as they show that the unique solution for the planning problem considering the utility of the existing agents (i.e., those who are alive just before the policy is introduced) corresponds to the $\mathcal{A}-$efficiency.

[47] The related work Zhou (2025) adopts the same welfare criteria in its steady-state analysis. In its transition analysis, Zhou (2025) also computes the average welfare of the existing households already alive before the policy changes to evaluate the $\mathcal{A}-$efficiency of the policy.

$$V^P_{j=1}(\boldsymbol{x}_1;\lambda) = \mathbb{E}\Bigg\{\sum_{j=1}^{J}\beta^{j-1}\cdot u(c_j\cdot(1+\lambda), l_j)$$
$$+\sum_{j=J_F}^{J_{IVT-1}}\beta^{j-1}b(n)\cdot v(q_j) + \beta^{J_{IVT}-1}\lambda_a\cdot b(n)\cdot V_{g0}(\boldsymbol{x}_{J_{IVT}})\Bigg\}$$

Here, $\{c_j, l_j, q_j, n, a_{IVT}\}$ are optimal choices with the policy P and each state $\boldsymbol{x}_j$. Finally, the consumption equivalence with a policy P is given as a scalar λ satisfying the following equation:

$$V^0(\lambda) = V^P(0).$$

That is, the consumption equivalence λ makes the newborn agents indifferent between the new economy and the benchmark by scaling up the benchmark consumption by λ.

G.2 Decomposing the Long-run Effects

The long-run changes in household decisions, such as college enrollment rates, can be broken down into direct effects and indirect effects. The direct effects capture the effects of introducing the policy holding macroeconomic objects—such as prices, tax rates, and distributions—fixed. The indirect effects capture the effects of the policy induced by changes in factor prices (i.e., GE effects), tax rates (i.e., taxation effects), and household distribution (i.e., distributional effects). To isolate each effect, I conduct a decomposition exercise as follows. I first solve an equilibrium by introducing the new grant function. Then, I solve the household problem by replacing one of the four equilibrium objects—-the grant function, prices, labor income tax rate, and household distribution—with its counterpart from the new equilibrium. Note that this method does not guarantee that each implied effect adds up to the overall effects because all factors except grant function are endogenous and interconnected. Table G.2 summarizes the decomposition results.

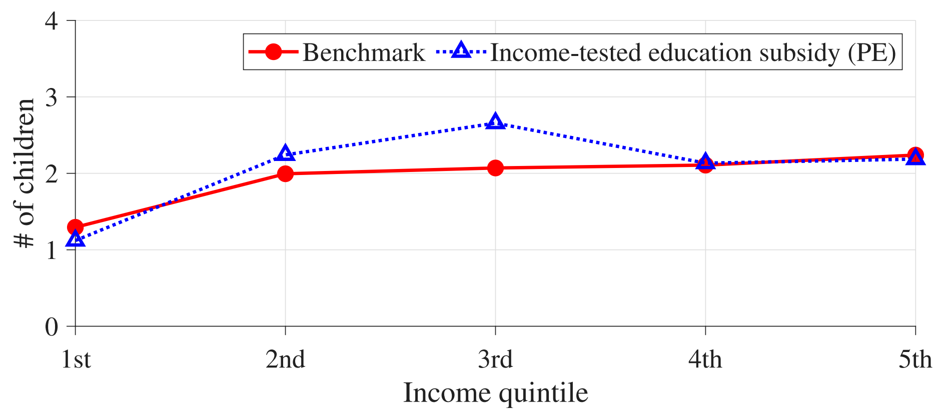
Table G.2: Decomposition of college enrollment increases

Outcome	Direct	GE	Tax	Dist.	All
College enrollment (Δ p.p.)	+4.1	−1.0	0.0	+2.0	+6.2
High-school-graduate parents	+4.4	−1.0	0.0	+1.5	+9.4
College-graduate parents	+2.6	−1.0	−1.0	+2.0	+1.4

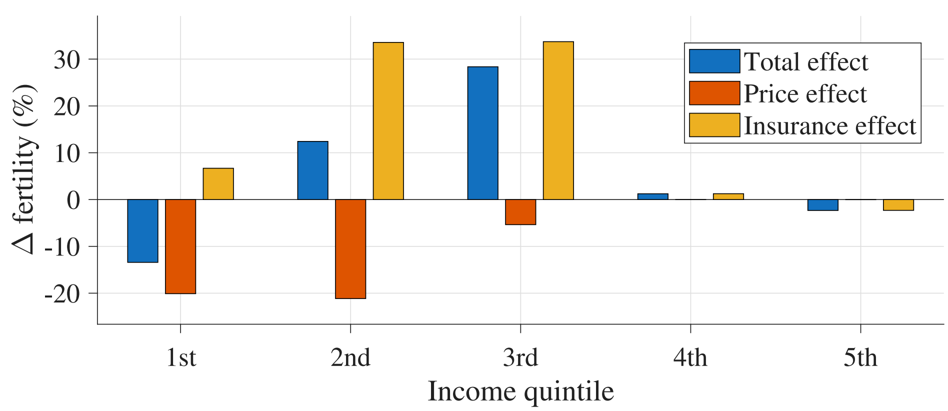
Note: Column "Direct" reports the direct effects of the grant policy, holding prices, tax rates, and distributions fixed. Columns "GE," "Tax," and "Dist." isolate the effects operating through equilibrium prices, labor income taxes, and changes in the household distribution, respectively. "All" reports the overall long-run effect of the policy.

For the long-run increase in the college enrollment rate, critical forces are direct and distributional effects. If the grant is introduced while other variables (i.e., prices, tax rate, and distribution) are fixed as in the benchmark, the college enrollment rate increases by 4.1 p.p., more than half of the overall increase. These direct effects capture the effects of relaxing financial constraints on college enrollment decisions. GE effects put downward

Figure G.7: Effects of education subsidies on fertility



(a) Fertility effects of education subsidies



(b) Decomposition across income quintiles pressure on college enrollment because the lower college premium reduces the incentive for college enrollment.

Note that these direct effects do not explain the overall effects, so we need other forces accounting for the increase in college enrollment rate; that turns out to be the distributional effects. The implied changes in the distribution in the long run, holding grant function, prices, and tax rate fixed as in the benchmark, lead to a 2.0 p.p. higher college enrollment rate. The most relevant factor is the change in the skill distribution of parents (i.e., the share of college graduates). The increase in the share of college graduates implies a higher share of those more likely to have children enrolling in college due to the intergenerational persistence of education. Then, the distributional change amplifies the short-run increase in college enrollment rate facilitated by the direct effects.

G.3 Expansion

Table G.3: Marginal expansion of the policy: an increase in the income threshold

	Income threshold	
	40th	60th
Education		
College enrollment (Δ p.p.)	+6.2	+5.1
Enrollment rate by parents' education (Δ p.p.)		
High-school graduates	+9.4	+5.0
College graduates	+1.4	−0.2
Inequality and mobility		
Average wage ratio (college/high school; Δ %)	+6.1	+8.6
Intergenerational persistence of income (Δ %)	−9.2	−8.1
Macroeconomic aggregates		
Output per capita (Δ %)	+1.3	−0.2
Capital (Δ %)	+0.0	−0.7
Labor (Δ %)	+1.9	+0.1
Labor income tax (Δ p.p.)	+0.11	+0.92
Welfare		
Consumption equivalence (%)	+6.4	−3.1

Note: The income threshold indicates the percentile of the household income distribution below which students are eligible for the grant. Fertility is endogenous in both setups. Welfare gains are reported as consumption-equivalent variations for newborn agents behind the veil of ignorance.